

CompTIA Linux+ (XK0-005)

Package Management and Updating Linux Devices

- Introduction
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 - Exercise 1 - Install and Update Software using Dnf
 - Exercise 2 - Install and Update Software using Apt
 - Review
-

Introduction

XK0-005

Dnf

Apt

Repository

Software Packages

Welcome to the **Package Management and Updating Linux Devices** Practice Lab. In this module, you will be provided with the instructions and devices needed to develop your hands-on skills.

Learning Outcomes

In this module, you will complete the following exercises:

- Exercise 1 - Install and Update Software using Dnf
- Exercise 2 - Install and Update Software using Apt

After completing this module, you should be able to:

- Use Dnf to Explore and Install a Current Repository
- Install a Custom Repository
- Remove an Installed Software Package
- Use Apt to Explore and Install a Current Repository
- Install a Third-party Repository
- Upgrade and Remove Installed Software

Exam Objectives

The following exam objectives are covered in this module:

1.6 Given a scenario, build and install software

- Package management

Lab Duration

It will take approximately **1 hour** to complete this lab.

Help and Support

For more information on using Practice Labs, please see our **Help and Support** page. You can also raise a technical support ticket from this page.

Click **Next** to view the Lab topology used in this module.

Lab Topology

During your session, you will have access to the following lab configuration.

Depending on the exercises, you may or may not use all of the devices, but they are shown here in the layout to get an overall understanding of the topology of the lab.

- **PLABSRV01** - Ubuntu 20.04 - Stand-Alone Ubuntu Server (Console)
- **PLABSRV02** - Alma Linux 8.5 - Stand-Alone Linux Server (GUI)
- **PLABCLIENT01** - Ubuntu 20.04 - Stand-Alone Ubuntu Workstation (GUI)
- **PLABWIN10** - Windows 10 - Stand-Alone Workstation

Click **Next** to proceed to the first exercise.

Exercise 1 - Install and Update Software using Dnf

A Linux repository contains important software for different types of Linux distributions. Every linux distribution has its own standard repository. Dnf is a software package manager that can perform installations, updates and removal of packages.

In this exercise, you will learn to manage software repositories using dnf in Alma Linux.

Learning Outcomes

After completing this exercise, you should be able to:

- Use Dnf to Explore and Install a Current Repository
- Install a Custom Repository
- Remove an Installed Software Package

Your Devices

You will be using the following device in this lab. Please power this on now.

- **PLABSRV02** - Alma Linux 8.5 - Stand-Alone Linux Server (GUI)

Task 1 - Use Dnf to Explore and Install a Current Repository

In this task, you will use dnf to explore and install the current repository.

Step 1

Connect to **PLABSRV02**.

Login as **admin** using the following **Password**:

Passw0rd

Press **Enter**.

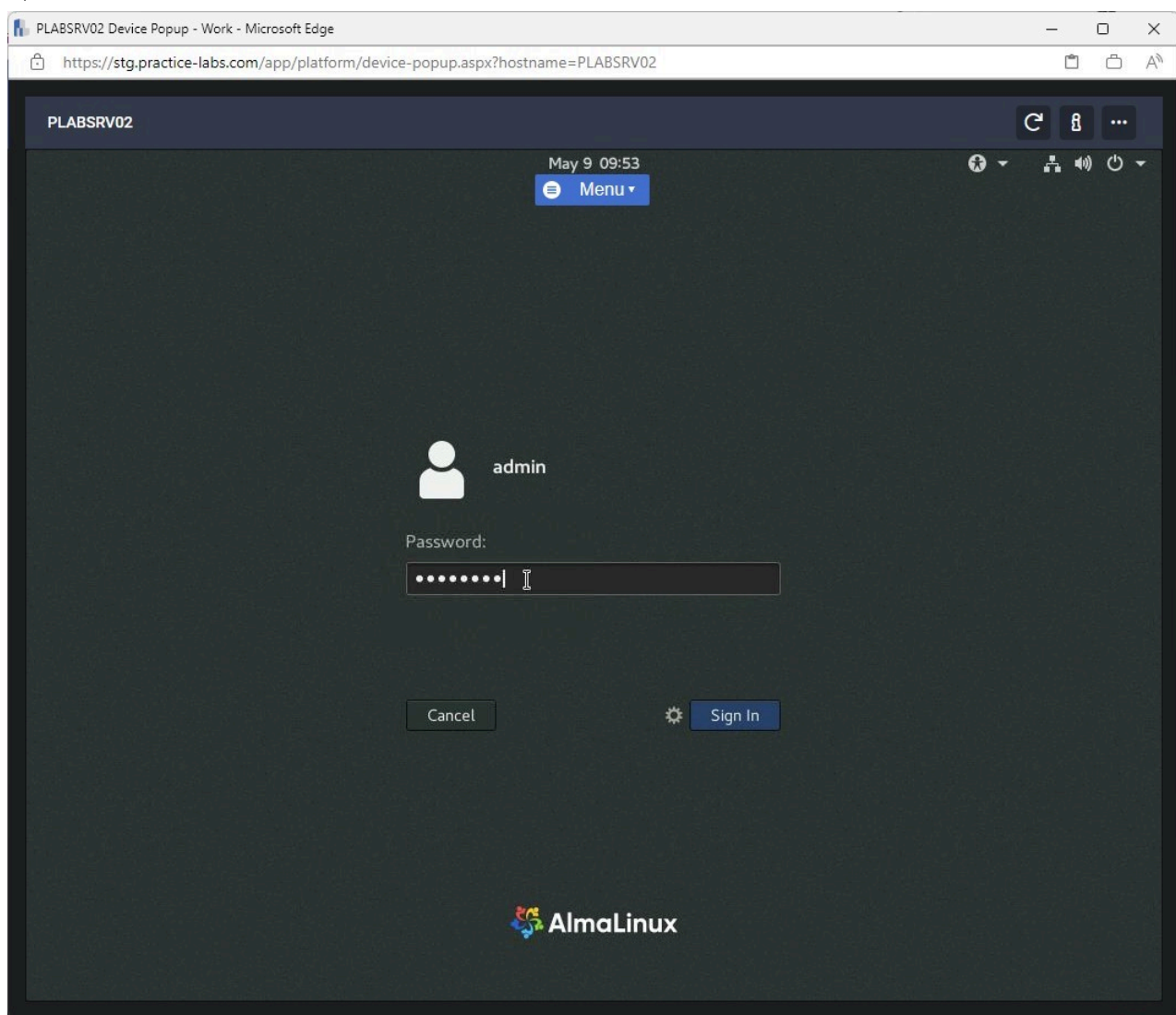


Figure 1.1 Screenshot of PLABSRV02: The login screen is displayed, showing the required Password typed in.

Step 2

Click **Activities** in the top left corner of the screen.

Select **Terminal** from the menu.

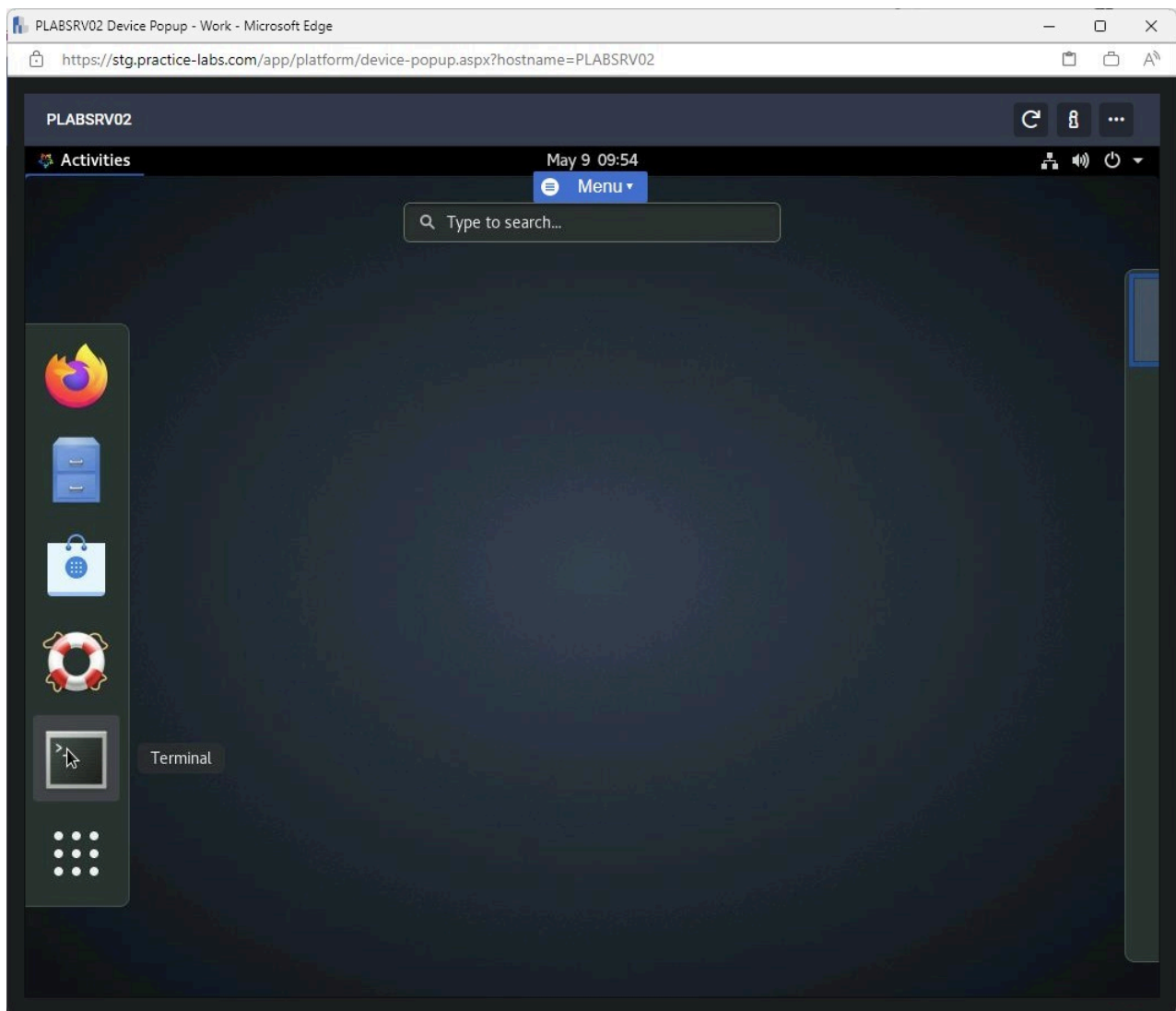


Figure 1.2 Screenshot of PLABSRV02: Activities menu is accessed, and Terminal is selected.

Step 3

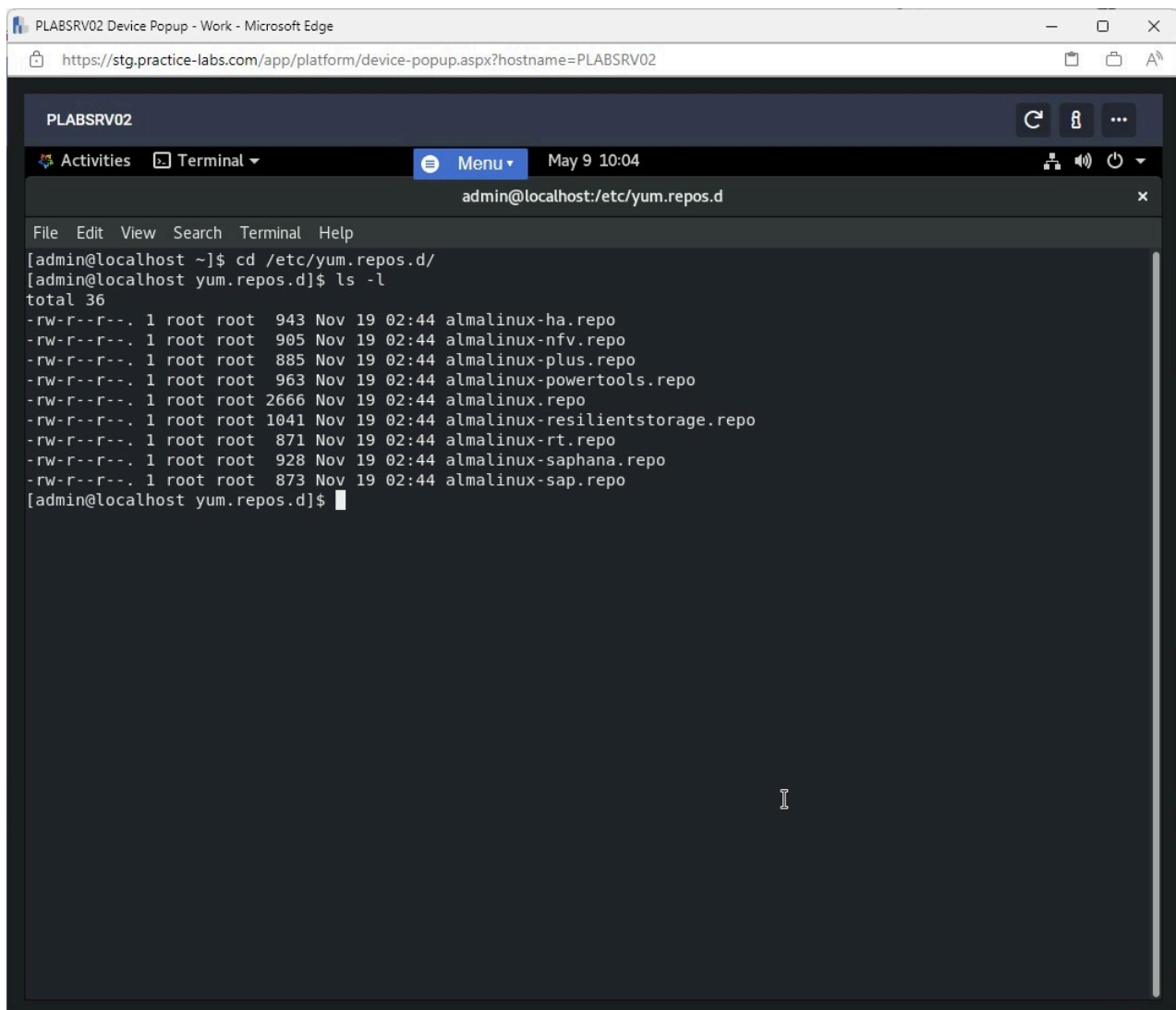
In the **Terminal** window, type the following command to navigate to the **yum.repos.d** repository:

```
cd /etc/yum.repos.d/
```

Press **Enter**.

Next, type the following command and press **Enter**:

```
ls -l
```



The screenshot shows a web browser window titled "PLABSRV02 Device Popup - Work - Microsoft Edge" with the URL "https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABSRV02". Inside the browser is a terminal window titled "PLABSRV02" with a menu bar (Activities, Terminal, Menu) and a clock showing "May 9 10:04". The terminal prompt is "admin@localhost:/etc/yum.repos.d". The user has entered the command "cd /etc/yum.repos.d/" and then "ls -l". The output shows a list of files in the directory, all with the ".repo" extension, including almalinux-ha.repo, almalinux-nfv.repo, almalinux-plus.repo, almalinux-powertools.repo, almalinux.repo, almalinux-resilientstorage.repo, almalinux-rt.repo, almalinux-saphana.repo, and almalinux-sap.repo. The terminal window has a menu bar (File, Edit, View, Search, Terminal, Help) and a scrollbar on the right.

```
admin@localhost:/etc/yum.repos.d$ cd /etc/yum.repos.d/
admin@localhost yum.repos.d$ ls -l
total 36
-rw-r--r--. 1 root root 943 Nov 19 02:44 almalinux-ha.repo
-rw-r--r--. 1 root root 905 Nov 19 02:44 almalinux-nfv.repo
-rw-r--r--. 1 root root 885 Nov 19 02:44 almalinux-plus.repo
-rw-r--r--. 1 root root 963 Nov 19 02:44 almalinux-powertools.repo
-rw-r--r--. 1 root root 2666 Nov 19 02:44 almalinux.repo
-rw-r--r--. 1 root root 1041 Nov 19 02:44 almalinux-resilientstorage.repo
-rw-r--r--. 1 root root 871 Nov 19 02:44 almalinux-rt.repo
-rw-r--r--. 1 root root 928 Nov 19 02:44 almalinux-saphana.repo
-rw-r--r--. 1 root root 873 Nov 19 02:44 almalinux-sap.repo
admin@localhost yum.repos.d$
```

Figure 1.3 Screenshot of PLABSRV02: The Terminal window is displayed, showing the required commands entered and executed.

Note: The yum repository is a repository manager. In Alma Linux, repositories are defined in the yum.repos.d repository. Notice all the files with the **.repo** extension.

Step 4

In the **Terminal** window, type the following command and press **Enter**:

```
nano almalinux.repo
```

The screenshot shows a web browser window titled "PLABSRV02 Device Popup - Work - Microsoft Edge" with the URL "https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABSRV02". Inside the browser, a terminal window is open, displaying the content of the `almalinux.repo` file. The terminal window has a title bar "admin@localhost:/etc/yum.repos.d" and a menu bar with "File", "Edit", "View", "Search", "Terminal", and "Help". The file content is as follows:

```

almalinux.repo

[baseos]
name=AlmaLinux $releasever - BaseOS
mirrorlist=https://mirrors.almalinux.org/mirrorlist/$releasever/baseos
# baseurl=https://repo.almalinux.org/almalinux/$releasever/BaseOS/$basearch/os/
enabled=1
gpgcheck=1
countme=1
gpgkey=file:///etc/pki/rpm-gpg/RPM-GPG-KEY-AlmaLinux

[appstream]
name=AlmaLinux $releasever - AppStream
mirrorlist=https://mirrors.almalinux.org/mirrorlist/$releasever/appstream
# baseurl=https://repo.almalinux.org/almalinux/$releasever/AppStream/$basearch/os/
enabled=1
gpgcheck=1
countme=1
gpgkey=file:///etc/pki/rpm-gpg/RPM-GPG-KEY-AlmaLinux

[extras]
name=AlmaLinux $releasever - Extras
mirrorlist=https://mirrors.almalinux.org/mirrorlist/$releasever/extras
# baseurl=https://repo.almalinux.org/almalinux/$releasever/extras/$basearch/os/
enabled=1
gpgcheck=1
countme=1
gpgkey=file:///etc/pki/rpm-gpg/RPM-GPG-KEY-AlmaLinux

## Sources
[baseos-source]
name=AlmaLinux $releasever - BaseOS Source

```

A red error message is visible at the bottom of the terminal window: "[File 'almalinux.repo' is unwritable]". The terminal window also shows a status bar with various keyboard shortcuts like `^G Get Help`, `^O Write Out`, `^W Where Is`, `^K Cut Text`, `^J Justify`, `^C Cur Pos`, `^M-U Undo`, `^X Exit`, `^R Read File`, `^N Replace`, `^U Uncut Text`, `^T To Spell`, `^G Go To Line`, and `^M-E Redo`.

Figure 1.4 Screenshot of PLABSRV02: The `almalinux.repo` file is displayed.

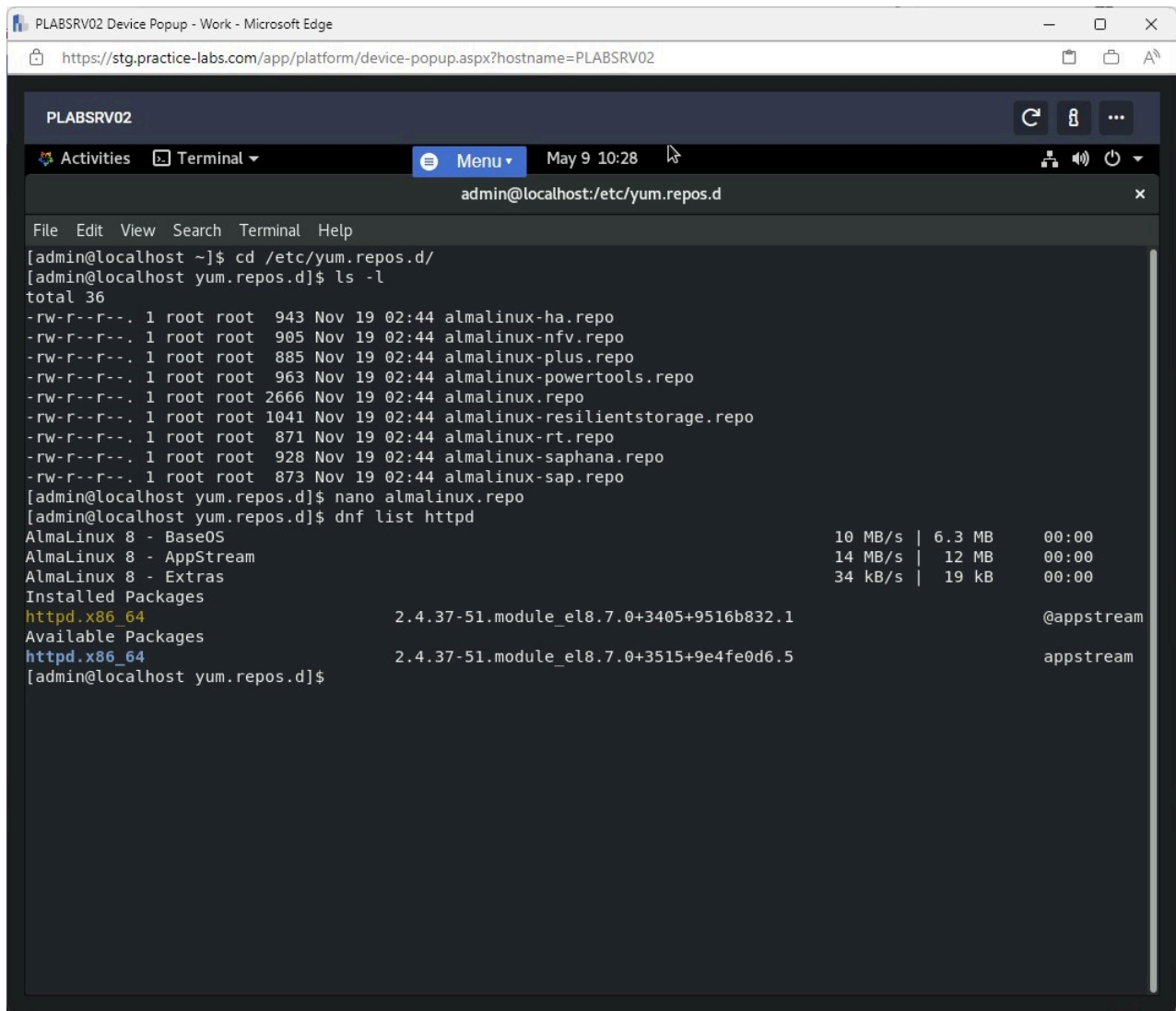
Note: The `almalinux.repo` file is displayed. Notice the **baseurl**, **mirrorlist** and **gpgkey**. The **baseurl** is the location of all the software in the repo. If unable to connect to the **baseurl**, the location in the **mirrorlist** will be searched. The **gpgkey** is a public cryptographic key. Though the current list is in the `yum.repos.d` directory, these repos will migrate to the `/etc/dnf/dnf.conf` directory.

Step 5

Press the **Ctrl+X** keys on the keyboard to exit the `almalinux.repo` file.

Back on the **Terminal** window, type the following command and press **Enter**:

```
dnf list httpd
```



```
PLABSRV02 Device Popup - Work - Microsoft Edge
https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABSRV02

PLABSRV02
Activities Terminal Menu May 9 10:28
admin@localhost:/etc/yum.repos.d

File Edit View Search Terminal Help
[admin@localhost ~]$ cd /etc/yum.repos.d/
[admin@localhost yum.repos.d]$ ls -l
total 36
-rw-r--r--. 1 root root 943 Nov 19 02:44 almalinux-ha.repo
-rw-r--r--. 1 root root 905 Nov 19 02:44 almalinux-nfv.repo
-rw-r--r--. 1 root root 885 Nov 19 02:44 almalinux-plus.repo
-rw-r--r--. 1 root root 963 Nov 19 02:44 almalinux-powertools.repo
-rw-r--r--. 1 root root 2666 Nov 19 02:44 almalinux.repo
-rw-r--r--. 1 root root 1041 Nov 19 02:44 almalinux-resilientstorage.repo
-rw-r--r--. 1 root root 871 Nov 19 02:44 almalinux-rt.repo
-rw-r--r--. 1 root root 928 Nov 19 02:44 almalinux-saphana.repo
-rw-r--r--. 1 root root 873 Nov 19 02:44 almalinux-sap.repo
[admin@localhost yum.repos.d]$ nano almalinux.repo
[admin@localhost yum.repos.d]$ dnf list httpd
AlmaLinux 8 - BaseOS                                10 MB/s | 6.3 MB    00:00
AlmaLinux 8 - AppStream                             14 MB/s | 12 MB    00:00
AlmaLinux 8 - Extras                                34 kB/s | 19 kB    00:00
Installed Packages
httpd.x86_64                                         2.4.37-51.module_el8.7.0+3405+9516b832.1 @appstream
Available Packages
httpd.x86_64                                         2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5 appstream
[admin@localhost yum.repos.d]$
```

Figure 1.5 Screenshot of PLABSRV02: The Terminal window is displayed, showing the required command entered and executed.

Note: The **dnf list** command is used to explore the contents of the **dnf** repository. When used with the **httpd** option, the command will list the **httpd** packages.

Step 6

In the **Terminal** window, type the following command and press **Enter**:

```
dnf search httpd
```

```

PLABSRV02 Device Popup - Work - Microsoft Edge
https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABSRV02

PLABSRV02
Activities Terminal Menu May 9 10:33
admin@localhost:/etc/yum.repos.d

File Edit View Search Terminal Help
-rw-r--r--. 1 root root 943 Nov 19 02:44 almalinux-ha.repo
-rw-r--r--. 1 root root 905 Nov 19 02:44 almalinux-nfv.repo
-rw-r--r--. 1 root root 885 Nov 19 02:44 almalinux-plus.repo
-rw-r--r--. 1 root root 963 Nov 19 02:44 almalinux-powertools.repo
-rw-r--r--. 1 root root 2666 Nov 19 02:44 almalinux.repo
-rw-r--r--. 1 root root 1041 Nov 19 02:44 almalinux-resilientstorage.repo
-rw-r--r--. 1 root root 871 Nov 19 02:44 almalinux-rt.repo
-rw-r--r--. 1 root root 928 Nov 19 02:44 almalinux-saphana.repo
-rw-r--r--. 1 root root 873 Nov 19 02:44 almalinux-sap.repo
[admin@localhost yum.repos.d]$ nano almalinux.repo
[admin@localhost yum.repos.d]$ dnf list httpd
AlmaLinux 8 - BaseOS                               10 MB/s | 6.3 MB   00:00
AlmaLinux 8 - AppStream                             14 MB/s | 12 MB   00:00
AlmaLinux 8 - Extras                                34 kB/s | 19 kB   00:00
Installed Packages
httpd.x86_64                                         2.4.37-51.module_el8.7.0+3405+9516b832.1 @appstream
Available Packages
httpd.x86_64                                         2.4.37-51.module_el8.7.0+3515+pe4fe0d6.5 appstream
[admin@localhost yum.repos.d]$ dnf search httpd
Last metadata expiration check: 0:04:44 ago on Tue 09 May 2023 10:28:18 AM BST.
===== Name Exactly Matched: httpd =====
httpd.x86_64 : Apache HTTP Server
===== Name & Summary Matched: httpd =====
almalinux-logos-httpd.noarch : AlmaLinux-related icons and pictures used by httpd
keycloak-httpd-client-install.noarch : Tools to configure Apache HTTPD as Keycloak client
python3-keycloak-httpd-client-install.noarch : Tools to configure Apache HTTPD as Keycloak client
===== Name Matched: httpd =====
httpd-devel.x86_64 : Development interfaces for the Apache HTTP server
httpdfilesystem.noarch : The basic directory layout for the Apache HTTP server
httpd-manual.noarch : Documentation for the Apache HTTP server
httpd-tools.x86_64 : Tools for use with the Apache HTTP Server
libmicrohttpd.i686 : Lightweight library for embedding a webserver in applications
libmicrohttpd.x86_64 : Lightweight library for embedding a webserver in applications
===== Summary Matched: httpd =====
mod_auth_mellon.x86_64 : A SAML 2.0 authentication module for the Apache Httpd Server
mod_dav_svn.x86_64 : Apache httpd module for Subversion server
[admin@localhost yum.repos.d]$

```

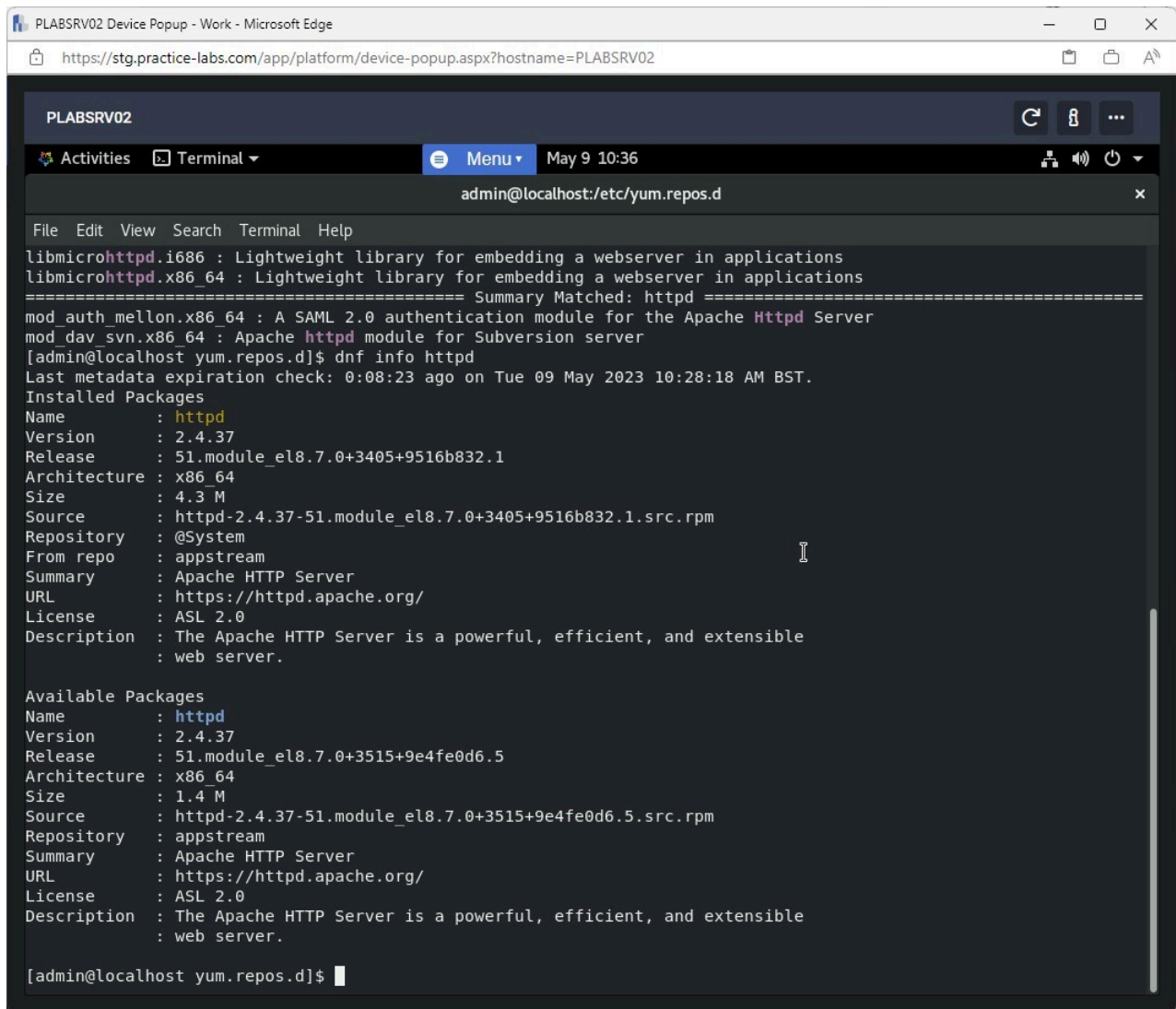
Figure 1.6 Screenshot of PLABSRV02: The Terminal window is displayed, showing the required command entered and executed.

Note: The *search* parameter will search by the keyword. Notice the matches for *httpd*.

Step 7

In the **Terminal** window, type the following command and press **Enter**:

```
dnf info httpd
```

The screenshot shows a web browser window titled "PLABSRV02 Device Popup - Work - Microsoft Edge" with the URL "https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABSRV02". Inside the browser, there is a terminal window titled "admin@localhost:/etc/yum.repos.d". The terminal displays the output of the command "dnf info httpd". The output shows information about the installed and available httpd packages, including their names, versions, releases, architectures, sizes, sources, repositories, summaries, URLs, licenses, and descriptions.

```
admin@localhost:/etc/yum.repos.d$ dnf info httpd
Last metadata expiration check: 0:08:23 ago on Tue 09 May 2023 10:28:18 AM BST.
Installed Packages
Name           : httpd
Version        : 2.4.37
Release       : 51.module_el8.7.0+3405+9516b832.1
Architecture  : x86_64
Size          : 4.3 M
Source        : httpd-2.4.37-51.module_el8.7.0+3405+9516b832.1.src.rpm
Repository    : @System
From repo     : appstream
Summary       : Apache HTTP Server
URL           : https://httpd.apache.org/
License       : ASL 2.0
Description    : The Apache HTTP Server is a powerful, efficient, and extensible
                  : web server.

Available Packages
Name           : httpd
Version        : 2.4.37
Release       : 51.module_el8.7.0+3515+9e4fe0d6.5
Architecture  : x86_64
Size          : 1.4 M
Source        : httpd-2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5.src.rpm
Repository    : appstream
Summary       : Apache HTTP Server
URL           : https://httpd.apache.org/
License       : ASL 2.0
Description    : The Apache HTTP Server is a powerful, efficient, and extensible
                  : web server.

[admin@localhost yum.repos.d]$
```

Figure 1.7 Screenshot of PLABSRV02: The Terminal window is displayed, showing the required command entered and executed.

Note: The *info* parameter will list all information on the **httpd** package.

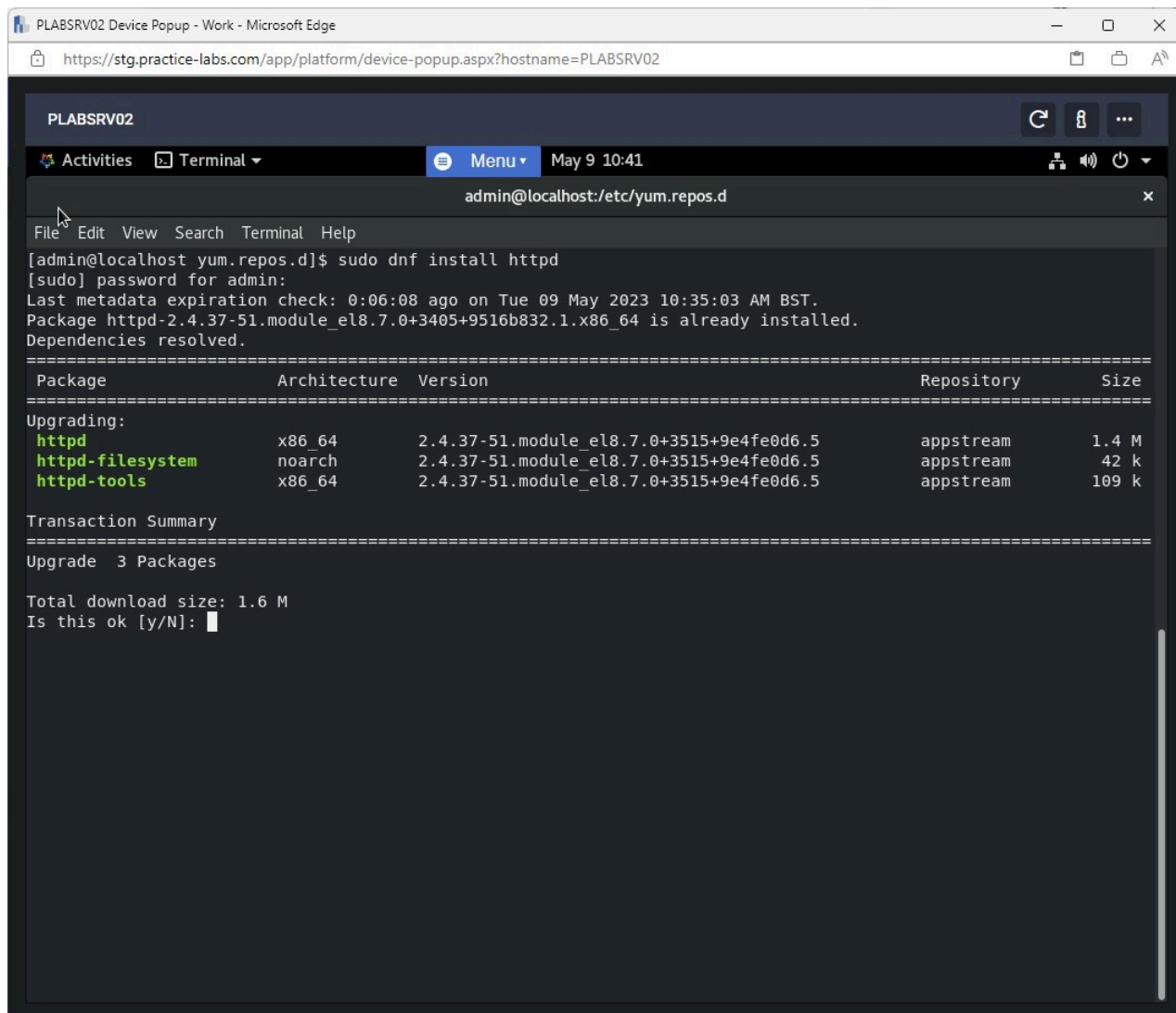
Step 8

In the **Terminal** window, type the following command and press **Enter**:

```
sudo dnf install httpd
```

Type the following **Password** and press **Enter**:

Passw0rd



The screenshot shows a web browser window titled "PLABSRV02 Device Popup - Work - Microsoft Edge" with the URL "https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABSRV02". Inside the browser, a terminal window is open, displaying the command "sudo dnf install httpd" and its output. The output shows that the package "httpd-2.4.37-51.module_el8.7.0+3405+9516b832.1.x86_64" is already installed, and dependencies are resolved. A table lists the packages to be upgraded: "httpd", "httpd-filesystem", and "httpd-tools". The transaction summary shows an upgrade of 3 packages with a total download size of 1.6 M. The prompt "Is this ok [y/N]:" is displayed at the bottom of the terminal.

```
admin@localhost:/etc/yum.repos.d$ sudo dnf install httpd
[sudo] password for admin:
Last metadata expiration check: 0:06:08 ago on Tue 09 May 2023 10:35:03 AM BST.
Package httpd-2.4.37-51.module_el8.7.0+3405+9516b832.1.x86_64 is already installed.
Dependencies resolved.
=====
Package                Architecture Version                                Repository    Size
=====
Upgrading:
httpd                   x86_64      2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5  appstream    1.4 M
httpd-filesystem        noarch      2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5  appstream    42 k
httpd-tools             x86_64      2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5  appstream    109 k
=====
Transaction Summary
=====
Upgrade 3 Packages

Total download size: 1.6 M
Is this ok [y/N]:
```

Figure 1.8 Screenshot of PLABSRV02: The Terminal window is displayed, showing the required command entered and executed.

Note: The **dnf install httpd** command will install the **httpd** software from a repository. Notice all that is installed. The application, the dependencies and weak dependencies.

Step 9

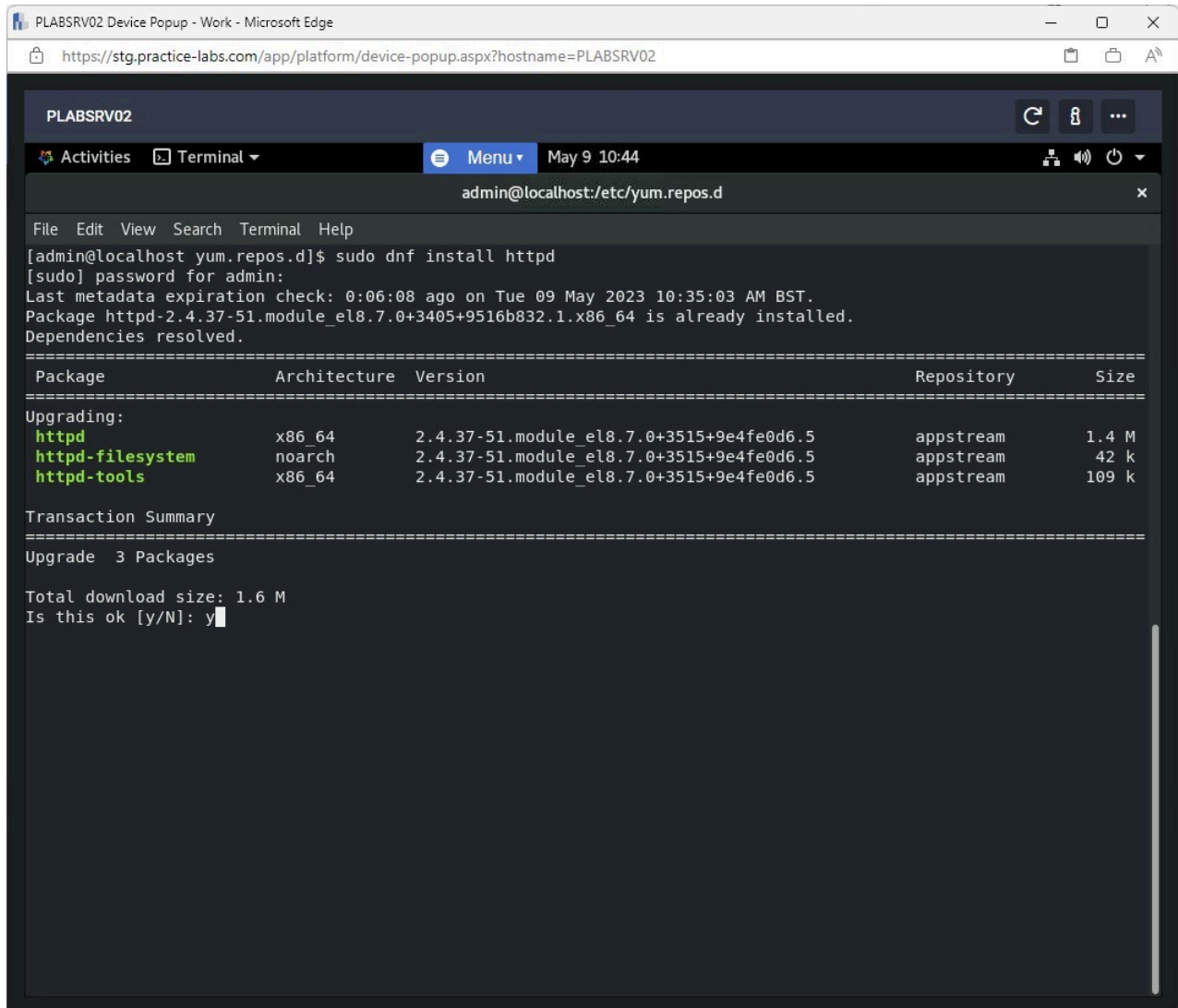
In the **Terminal** window, type the following for the **Is this ok [y/N]** prompt:

y

Press **Enter**.

Note: If you encounter an error when attempting to install `httpd`, run the following command and then re-try, this is to update the GPG key required to download and install packages :

```
sudo rpm --import https://repo.almalinux.org/almalinux/RPM-GPG-KEY-AlmaLinux
```



The screenshot shows a terminal window titled "admin@localhost:/etc/yum.repos.d" within a web browser. The terminal displays the command `sudo dnf install httpd` and its output. It shows that the `httpd` package is already installed and lists the packages to be upgraded: `httpd`, `httpd-filesystem`, and `httpd-tools`. The transaction summary indicates an upgrade of 3 packages with a total download size of 1.6 M. The prompt "Is this ok [y/N]: y" is shown with the letter 'y' entered.

```
admin@localhost:/etc/yum.repos.d$ sudo dnf install httpd
[sudo] password for admin:
Last metadata expiration check: 0:06:08 ago on Tue 09 May 2023 10:35:03 AM BST.
Package httpd-2.4.37-51.module_el8.7.0+3405+9516b832.1.x86_64 is already installed.
Dependencies resolved.
=====
Package                Architecture Version                                Repository    Size
=====
Upgrading:
httpd                   x86_64      2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5  appstream    1.4 M
httpd-filesystem        noarch      2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5  appstream     42 k
httpd-tools             x86_64      2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5  appstream    109 k
=====
Transaction Summary
=====
Upgrade 3 Packages

Total download size: 1.6 M
Is this ok [y/N]: y
```

Figure 1.9 Screenshot of PLABSRV02: The Terminal window is displayed, showing y entered for the Is this ok [y/N] prompt.

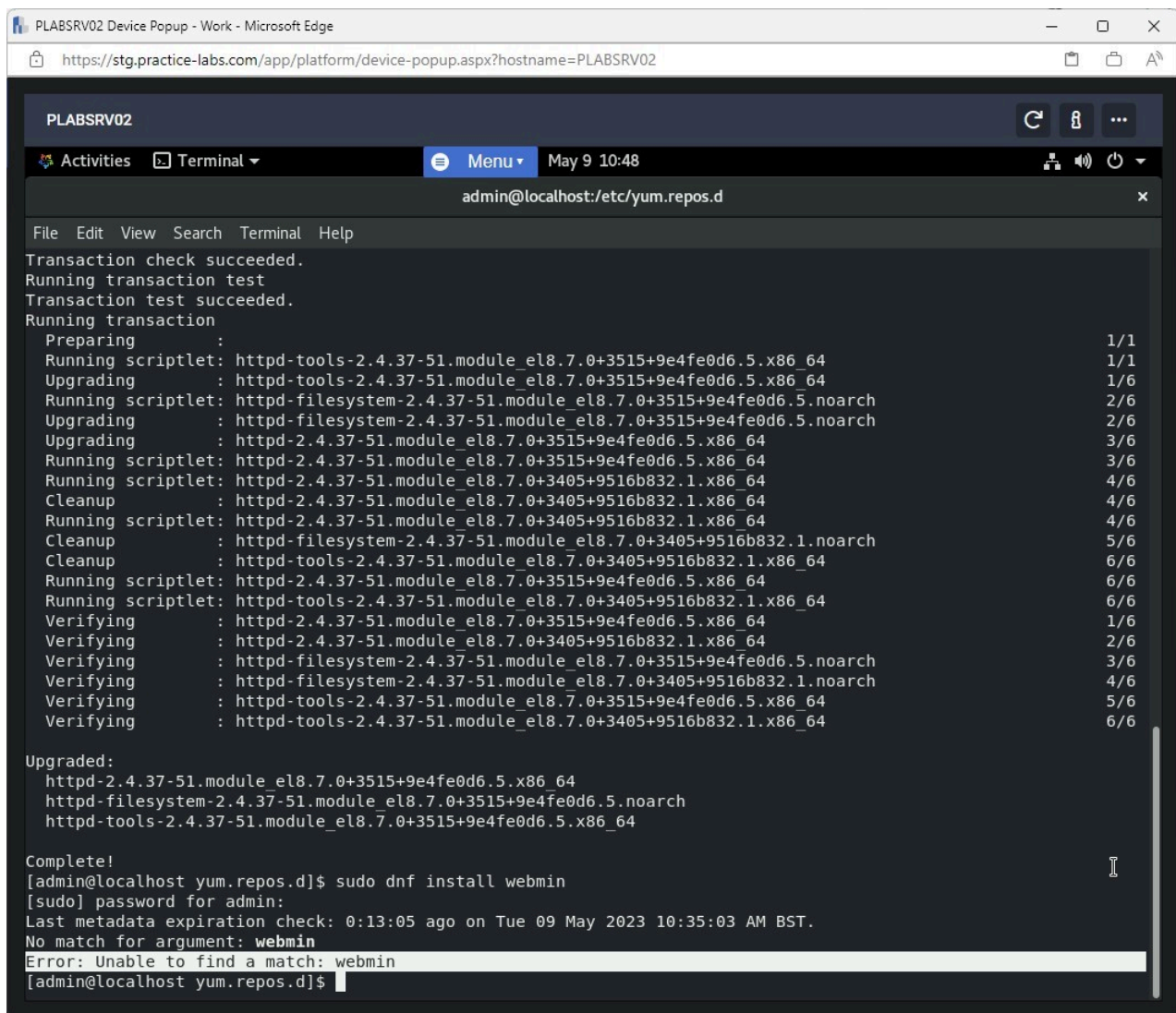
Step 10

Once the installation is complete, type the following command and press **Enter**:


```
sudo dnf install webmin
```

Type the following **Password** and press **Enter**:

Passw0rd



```
PLABSRV02 Device Popup - Work - Microsoft Edge
https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABSRV02

PLABSRV02
Activities Terminal Menu May 9 10:48
admin@localhost:/etc/yum.repos.d

File Edit View Search Terminal Help
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing : 1/1
  Running scriptlet: httpd-tools-2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5.x86_64 1/1
  Upgrading : httpd-tools-2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5.x86_64 1/6
  Running scriptlet: httpd-filesystem-2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5.noarch 2/6
  Upgrading : httpd-filesystem-2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5.noarch 2/6
  Upgrading : httpd-2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5.x86_64 3/6
  Running scriptlet: httpd-2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5.x86_64 3/6
  Running scriptlet: httpd-2.4.37-51.module_el8.7.0+3405+9516b832.1.x86_64 4/6
  Cleanup : httpd-2.4.37-51.module_el8.7.0+3405+9516b832.1.x86_64 4/6
  Running scriptlet: httpd-2.4.37-51.module_el8.7.0+3405+9516b832.1.x86_64 4/6
  Cleanup : httpd-filesystem-2.4.37-51.module_el8.7.0+3405+9516b832.1.noarch 5/6
  Cleanup : httpd-tools-2.4.37-51.module_el8.7.0+3405+9516b832.1.x86_64 6/6
  Running scriptlet: httpd-2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5.x86_64 6/6
  Running scriptlet: httpd-tools-2.4.37-51.module_el8.7.0+3405+9516b832.1.x86_64 6/6
  Verifying : httpd-2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5.x86_64 1/6
  Verifying : httpd-2.4.37-51.module_el8.7.0+3405+9516b832.1.x86_64 2/6
  Verifying : httpd-filesystem-2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5.noarch 3/6
  Verifying : httpd-filesystem-2.4.37-51.module_el8.7.0+3405+9516b832.1.noarch 4/6
  Verifying : httpd-tools-2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5.x86_64 5/6
  Verifying : httpd-tools-2.4.37-51.module_el8.7.0+3405+9516b832.1.x86_64 6/6

Upgraded:
  httpd-2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5.x86_64
  httpd-filesystem-2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5.noarch
  httpd-tools-2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5.x86_64

Complete!
[admin@localhost yum.repos.d]$ sudo dnf install webmin
[sudo] password for admin:
Last metadata expiration check: 0:13:05 ago on Tue 09 May 2023 10:35:03 AM BST.
No match for argument: webmin
Error: Unable to find a match: webmin
[admin@localhost yum.repos.d]$
```

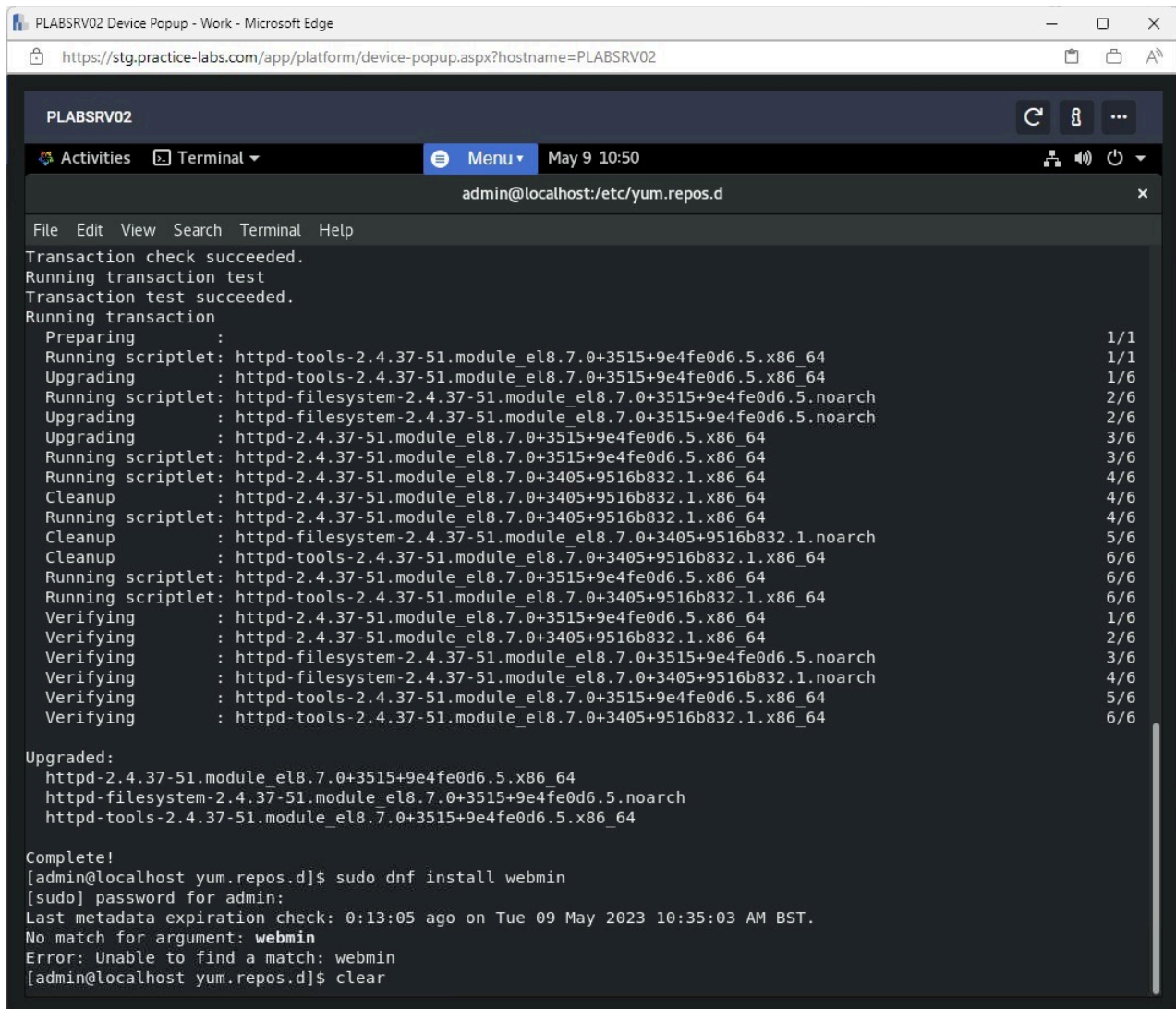
Figure 1.10 Screenshot of PLABSRV02: The Terminal window is displayed, showing the required command entered and executed.

Note: Notice no matches were found for the **webmin** software in the **dnf** repository.

Step 11

In the **Terminal** window, type the following command and press **Enter**:

clear



```
PLABSRV02 Device Popup - Work - Microsoft Edge
https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABSRV02

PLABSRV02
Activities Terminal Menu May 9 10:50
admin@localhost:/etc/yum.repos.d

File Edit View Search Terminal Help
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing : 1/1
  Running scriptlet: httpd-tools-2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5.x86_64 1/1
  Upgrading : httpd-tools-2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5.x86_64 1/6
  Running scriptlet: httpd-filesystem-2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5.noarch 2/6
  Upgrading : httpd-filesystem-2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5.noarch 2/6
  Upgrading : httpd-2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5.x86_64 3/6
  Running scriptlet: httpd-2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5.x86_64 3/6
  Running scriptlet: httpd-2.4.37-51.module_el8.7.0+3405+9516b832.1.x86_64 4/6
  Cleanup : httpd-2.4.37-51.module_el8.7.0+3405+9516b832.1.x86_64 4/6
  Running scriptlet: httpd-2.4.37-51.module_el8.7.0+3405+9516b832.1.x86_64 4/6
  Cleanup : httpd-filesystem-2.4.37-51.module_el8.7.0+3405+9516b832.1.noarch 5/6
  Cleanup : httpd-tools-2.4.37-51.module_el8.7.0+3405+9516b832.1.x86_64 6/6
  Running scriptlet: httpd-2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5.x86_64 6/6
  Running scriptlet: httpd-tools-2.4.37-51.module_el8.7.0+3405+9516b832.1.x86_64 6/6
  Verifying : httpd-2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5.x86_64 1/6
  Verifying : httpd-2.4.37-51.module_el8.7.0+3405+9516b832.1.x86_64 2/6
  Verifying : httpd-filesystem-2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5.noarch 3/6
  Verifying : httpd-filesystem-2.4.37-51.module_el8.7.0+3405+9516b832.1.noarch 4/6
  Verifying : httpd-tools-2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5.x86_64 5/6
  Verifying : httpd-tools-2.4.37-51.module_el8.7.0+3405+9516b832.1.x86_64 6/6

Upgraded:
  httpd-2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5.x86_64
  httpd-filesystem-2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5.noarch
  httpd-tools-2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5.x86_64

Complete!
[admin@localhost yum.repos.d]$ sudo dnf install webmin
[sudo] password for admin:
Last metadata expiration check: 0:13:05 ago on Tue 09 May 2023 10:35:03 AM BST.
No match for argument: webmin
Error: Unable to find a match: webmin
[admin@localhost yum.repos.d]$ clear
```

Figure 1.11 Screenshot of PLABSRV02: The Terminal window is displayed, showing the required command entered and executed.

Note: The *clear* command will clear the Terminal window.

Keep the **Terminal** window open.

Task 2 - Install a Custom Repository

In this task, you will install a custom repository named **webmin.repo** to install software not found in the default repository.

Step 1

Connect to **PLABSRV02**, where the **Terminal** window is open.

In the **Terminal** window, type the following command and press **Enter**:

```
sudoedit /etc/yum.repos.d/webmin.repo
```

Type the following **Password** and press **Enter**:

Passw0rd

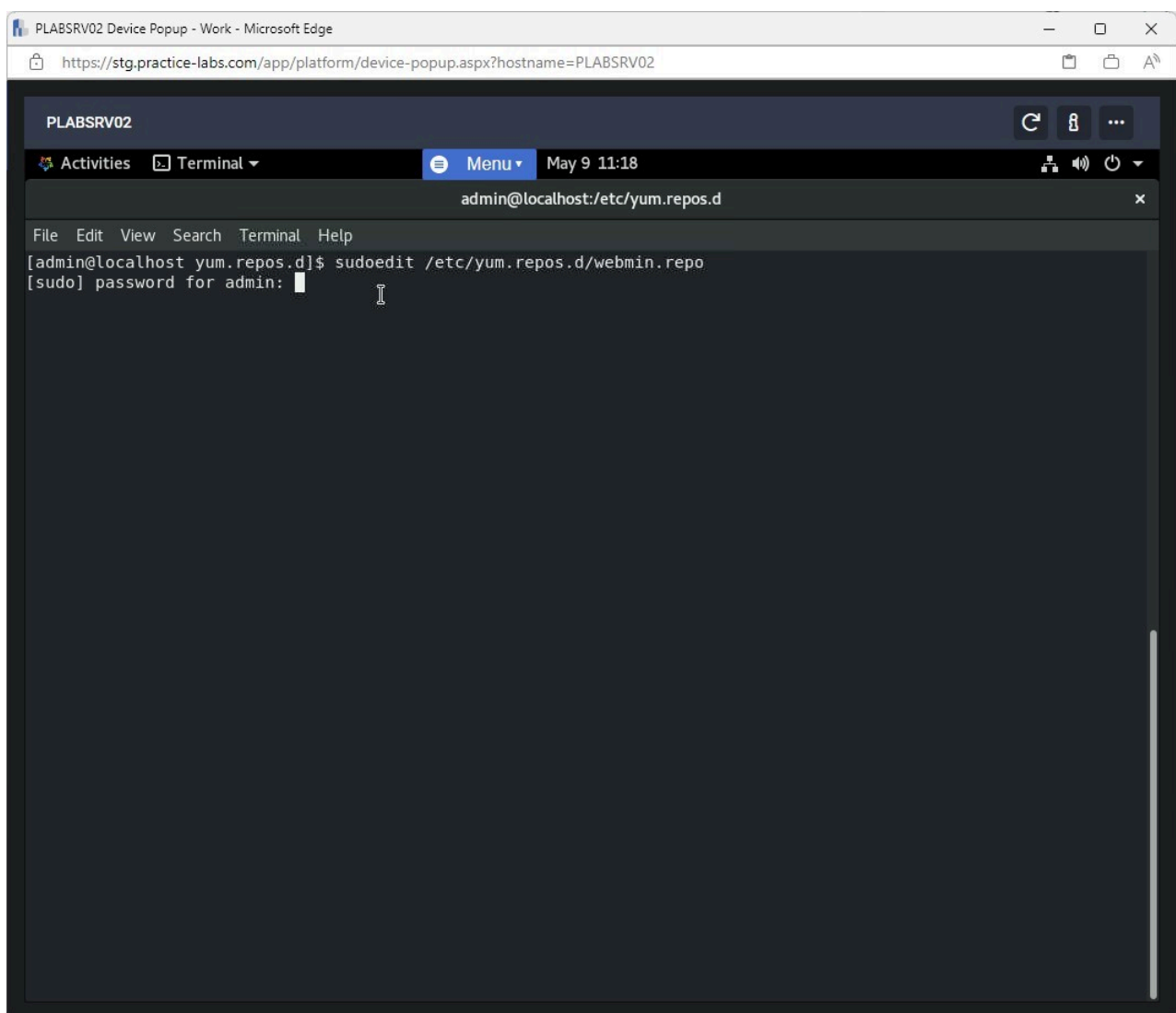


Figure 1.12 Screenshot of PLABSRV02: The Terminal window is displayed, showing the required command entered.

Note: You will add a custom repository in the `yum.repos.d` file.

Step 2

In the **yum.repos.d** file, press the **Esc** key and then the **I** key on the keyboard to enter **Insert** mode.

Next, type the following:

```
[Webmin]
name=Webmin Distribution Neutral
#baseurl=https://download.webmin.com/download/yum
mirrorlist=https://download.webmin.com/download/yum/
mirrorlist
enabled=1
```

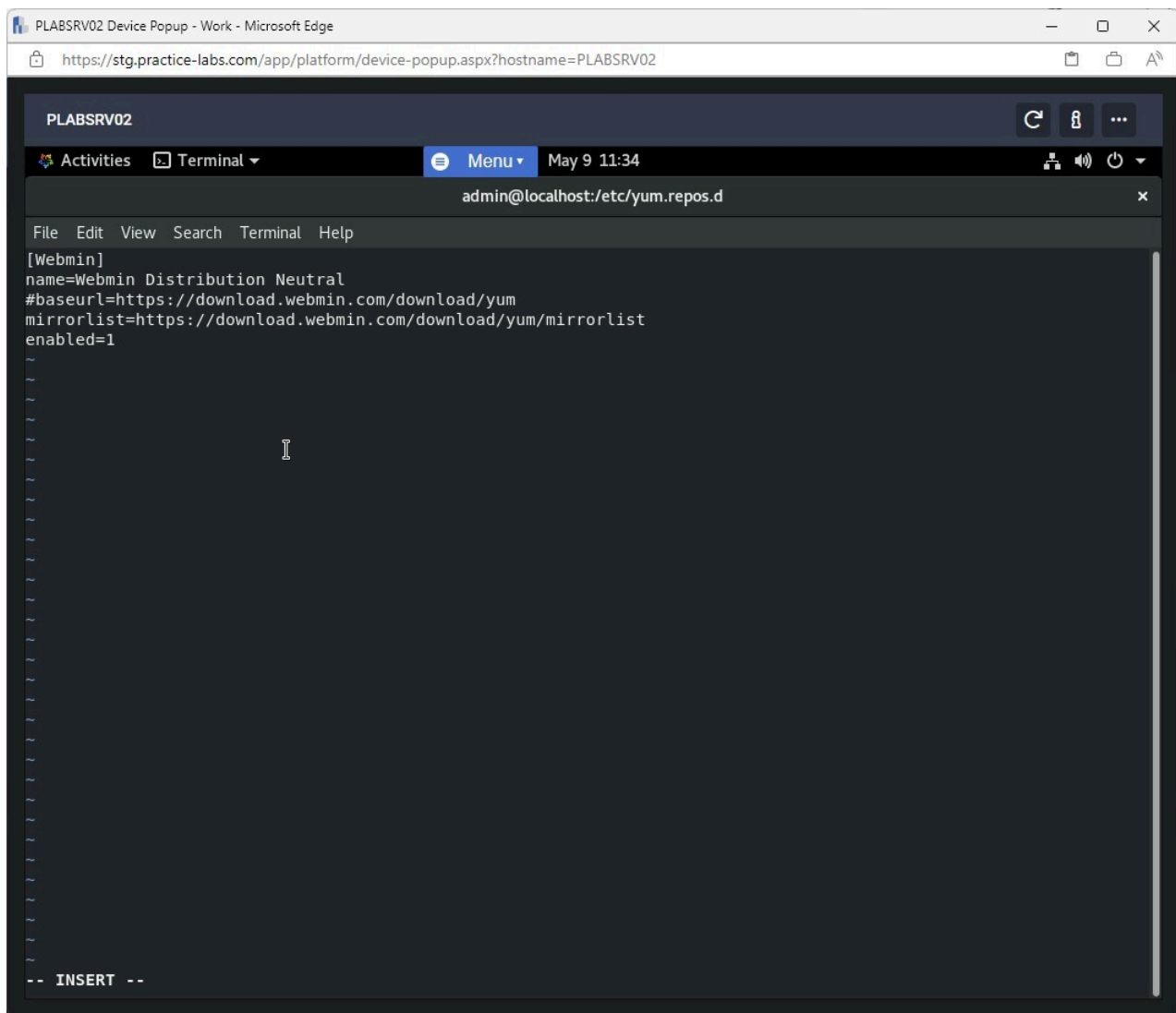


Figure 1.13 Screenshot of PLABSRV02: The yum.repos.d text editor window is displayed, showing the required text entered.

Step 3

Press the **Esc** key on the keyboard.

Next, type the following to exit the text editor window:

```
:wq
```

Press **Enter**.

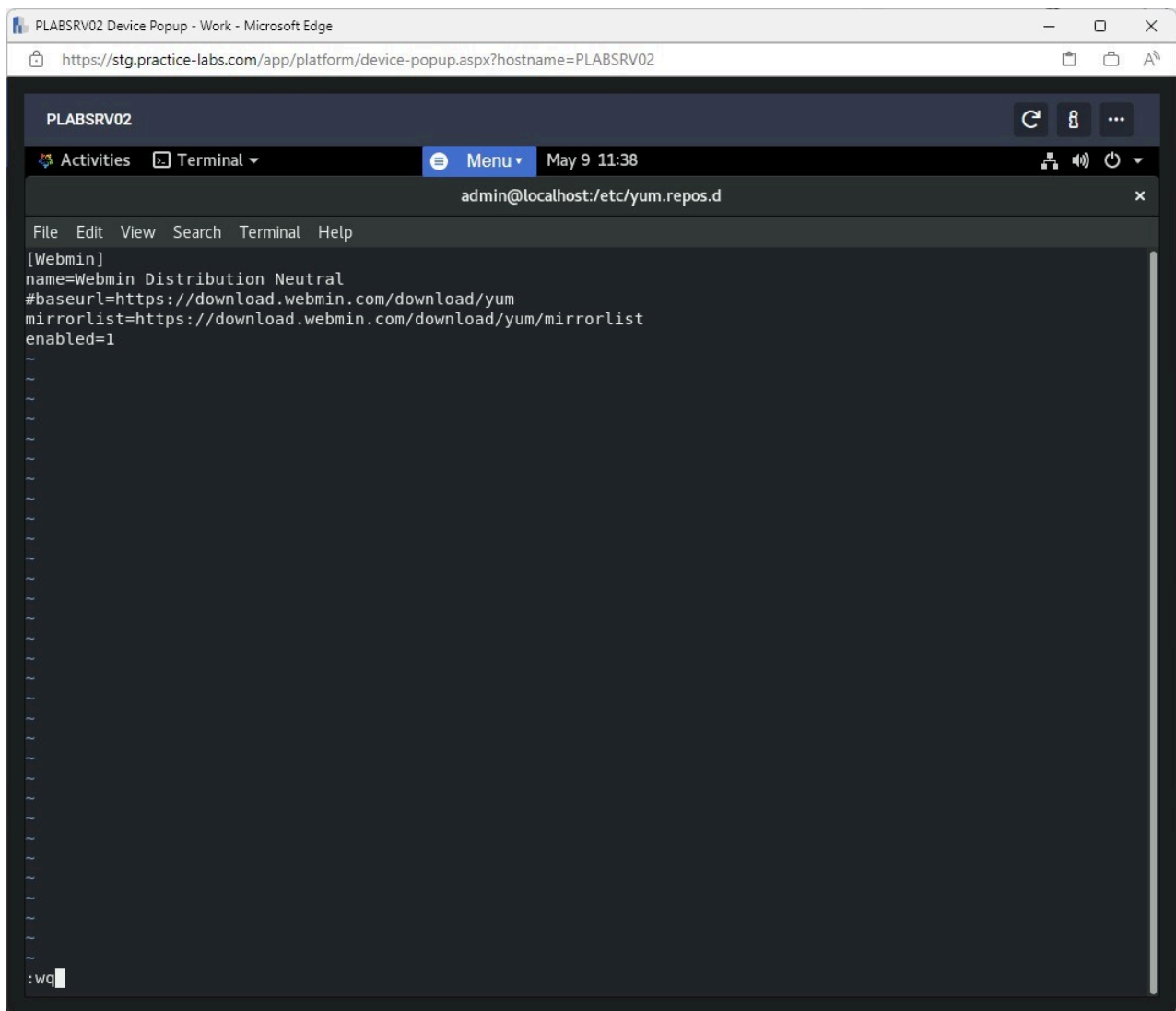


Figure 1.14 Screenshot of PLABSRV02: The yum.repos.d text editor window is displayed, showing the required text entered.

Step 4

You will now import the signing (**gpgkey**) key.

In the **Terminal** window, type the following command and press **Enter**:

```
sudo wget https://www.webmin.com/jcameron-key.asc
```

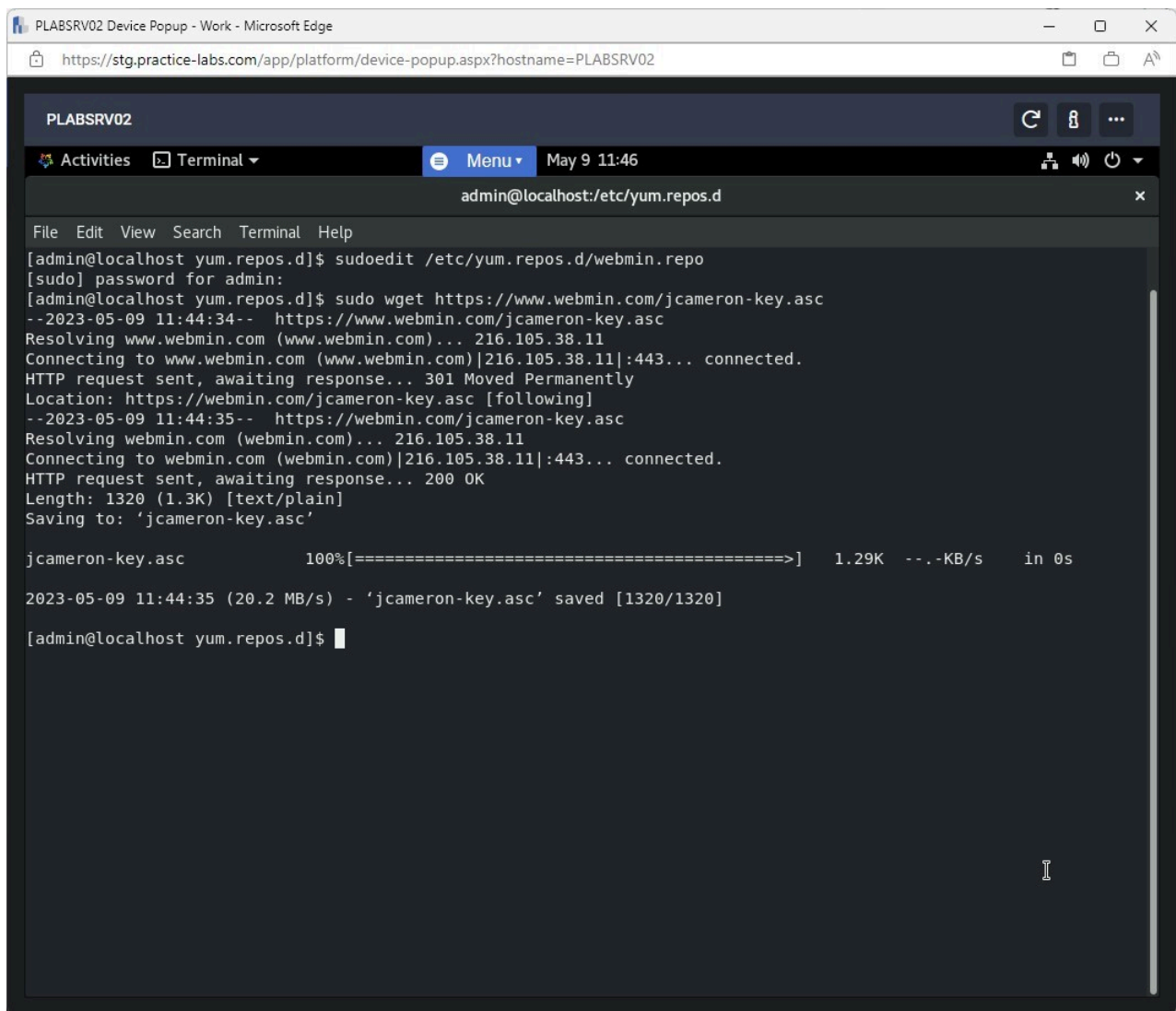



Figure 1.15 Screenshot of PLABSRV02: The Terminal window is displayed, showing the required command entered and executed.

Step 5

In the **Terminal** window, type the following command and press **Enter**:

```
sudo rpm -import jcameron-key.asc
```

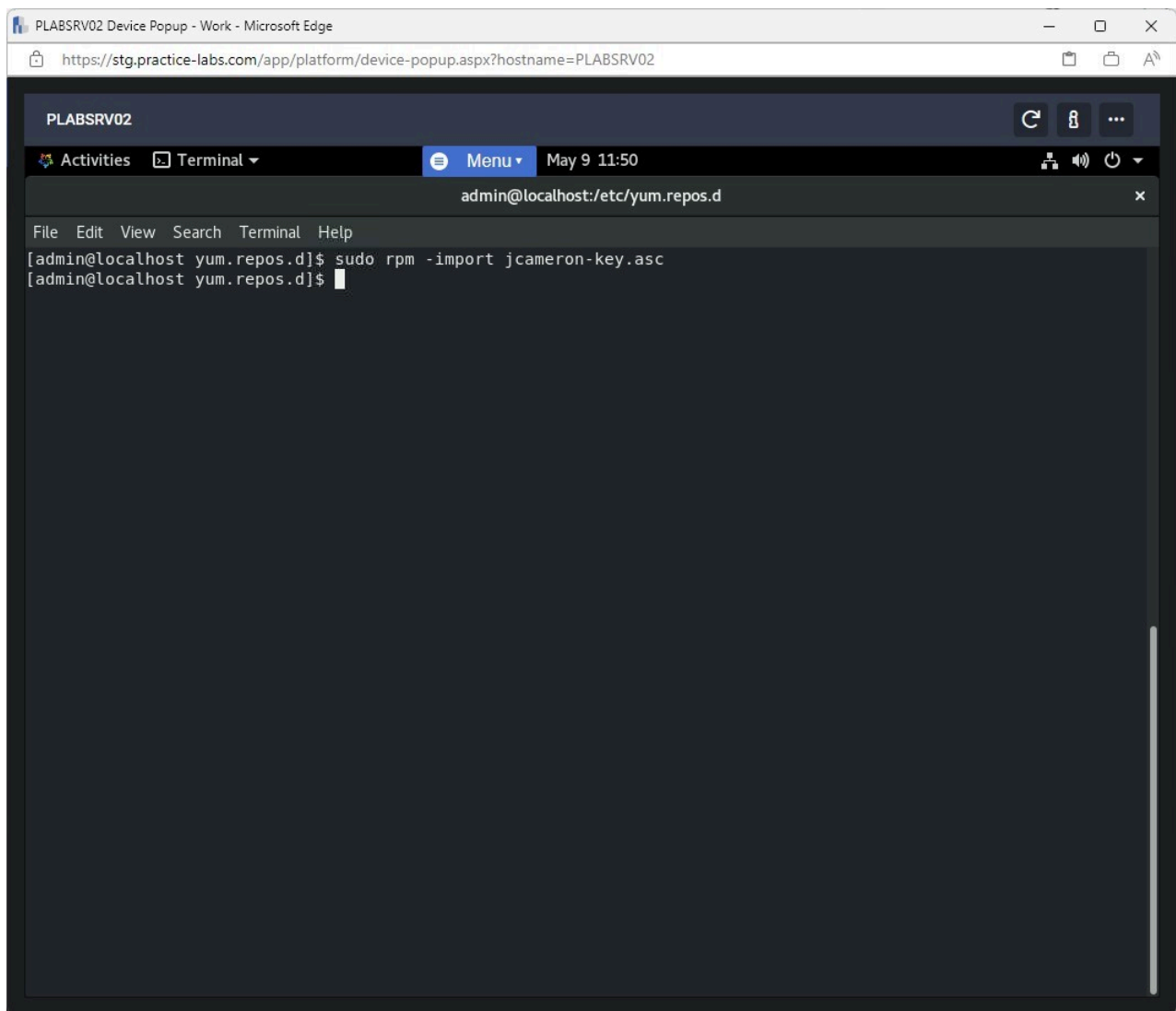


Figure 1.16 Screenshot of PLABSRV02: The Terminal window is displayed, showing the required command entered and executed.

Step 6

In the **Terminal** window, type the following command and press **Enter**:

```
sudo dnf install webmin
```

Next, type the following for the **Is this ok [y/N]** prompt:

```
y
```

Press **Enter**.

The screenshot shows a web browser window titled "PLABSRV02 Device Popup - Work - Microsoft Edge" with the URL "https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABSRV02". The browser displays a terminal window for "admin@localhost:/etc/yum.repos.d". The terminal shows the following commands and output:

```

[admin@localhost yum.repos.d]$ sudo rpm -import jcameron-key.asc
[admin@localhost yum.repos.d]$ sudo dnf install webmin
Webmin Distribution Neutral
Last metadata expiration check: 0:00:09 ago on Tue 09 May 2023 11:52:26 AM BST.
Dependencies resolved.

```

Package	Architecture	Version	Repository	Size
Installing:				
webmin	noarch	2.021-1	Webmin	40 M
Installing dependencies:				
perl-Digest-SHA	x86_64	1:6.02-1.el8	appstream	66 k
perl-Encode-Detect	x86_64	1.01-28.el8	appstream	90 k
perl-Filter	x86_64	2:1.58-2.el8	appstream	82 k
perl-encoding	x86_64	4:2.22-3.el8	appstream	68 k
perl-open	noarch	1.11-421.el8	appstream	77 k

```

Transaction Summary
-----
Install 6 Packages

Total download size: 40 M
Installed size: 116 M
Is this ok [y/N]: y

```

Figure 1.17 Screenshot of PLABSRV02: The Terminal window is displayed, showing the required command entered.

Step 7

The custom repository **webmin.repo** is now successfully installed.

```

PLABSRV02 Device Popup - Work - Microsoft Edge
https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABSRV02

PLABSRV02
Activities Terminal Menu May 9 11:55
admin@localhost:/etc/yum.repos.d

File Edit View Search Terminal Help
Is this ok [y/N]: y
Downloading Packages:
(1/6): perl-Digest-SHA-1:6.02-1.el8.x86_64.rpm 1.3 MB/s | 66 kB 00:00
(2/6): perl-Encode-Detect-1.01-28.el8.x86_64.rpm 1.6 MB/s | 90 kB 00:00
(3/6): perl-encoding-4:2.22-3.el8.x86_64.rpm 3.1 MB/s | 68 kB 00:00
(4/6): perl-Filter-2:1.58-2.el8.x86_64.rpm 1.1 MB/s | 82 kB 00:00
(5/6): perl-open-1.11-421.el8.noarch.rpm 1.1 MB/s | 77 kB 00:00
(6/6): webmin-2.021-1.noarch.rpm 3.3 MB/s | 40 MB 00:11
-----
Total 3.1 MB/s | 40 MB 00:12
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing                : 1/1
  Installing               : perl-Filter-2:1.58-2.el8.x86_64 1/6
  Installing               : perl-encoding-4:2.22-3.el8.x86_64 2/6
  Installing               : perl-open-1.11-421.el8.noarch 3/6
  Installing               : perl-Encode-Detect-1.01-28.el8.x86_64 4/6
  Installing               : perl-Digest-SHA-1:6.02-1.el8.x86_64 5/6
  Running scriptlet: webmin-2.021-1.noarch 6/6
  Installing               : webmin-2.021-1.noarch 6/6
  Running scriptlet: webmin-2.021-1.noarch 6/6
  Verifying              : perl-Digest-SHA-1:6.02-1.el8.x86_64 1/6
  Verifying              : perl-Encode-Detect-1.01-28.el8.x86_64 2/6
  Verifying              : perl-Filter-2:1.58-2.el8.x86_64 3/6
  Verifying              : perl-encoding-4:2.22-3.el8.x86_64 4/6
  Verifying              : perl-open-1.11-421.el8.noarch 5/6
  Verifying              : webmin-2.021-1.noarch 6/6

Installed:
perl-Digest-SHA-1:6.02-1.el8.x86_64 perl-Encode-Detect-1.01-28.el8.x86_64 perl-Filter-2:1.58-2.el8.x86_64
perl-encoding-4:2.22-3.el8.x86_64 perl-open-1.11-421.el8.noarch webmin-2.021-1.noarch

Complete!
[admin@localhost yum.repos.d]$

```

Figure 1.18 Screenshot of PLABSRV02: The Terminal window is displayed, showing the successful installation.

Note: The installation can take a few minutes to complete.

Keep the **Terminal** window open.

Task 3 - Remove an Installed Software Package

In this task, you will use `dnf` to remove an installed software package.

Step 1

In **PLABSRV02**, the **Terminal** window is open.

In the **Terminal** window, type the following command and press **Enter**:

```
sudo dnf remove httpd
```

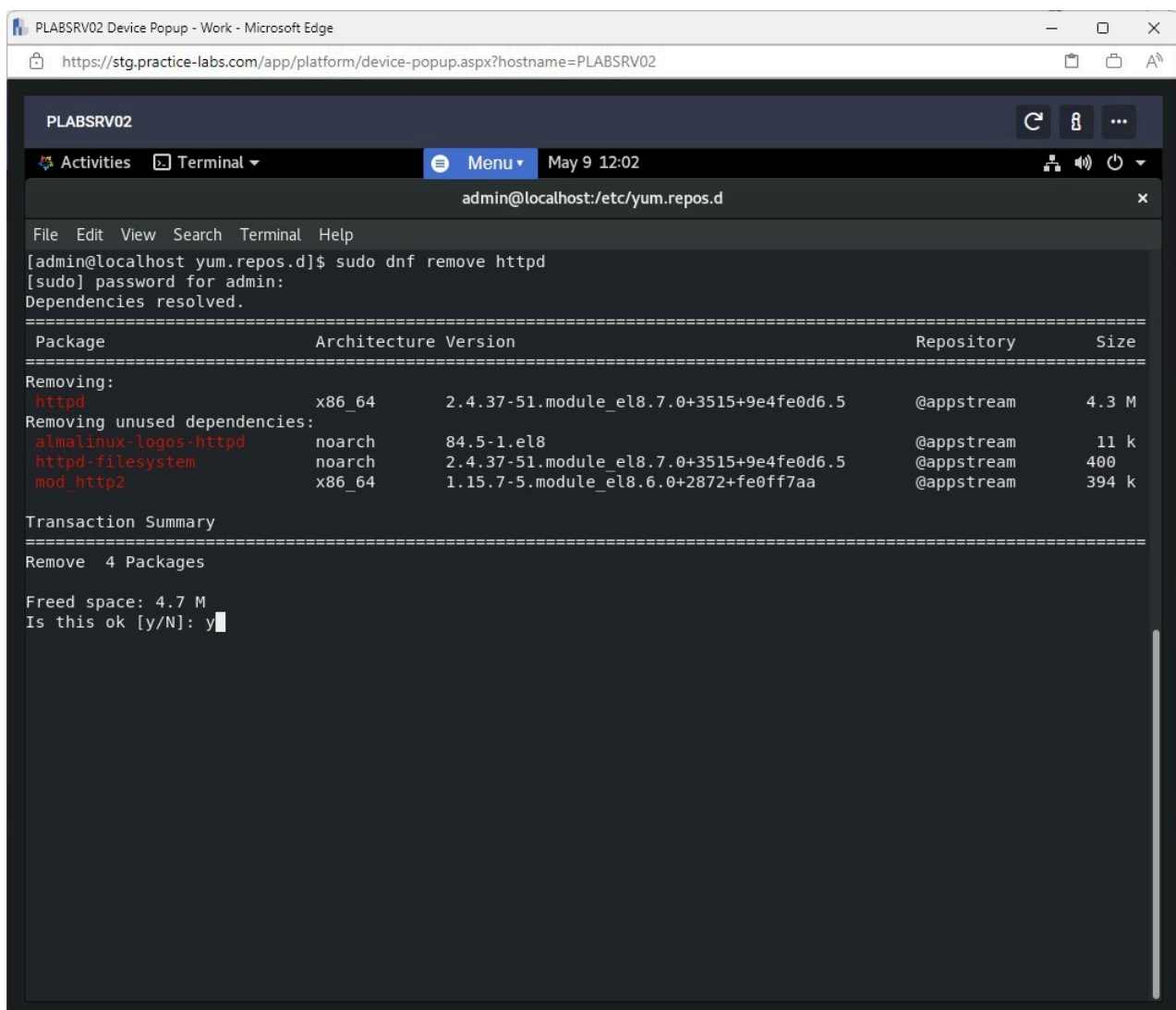
Type the following **Password** and press **Enter**:

Passw0rd

Next, type the following for the **Is this ok [y/N]** prompt:

y

Press **Enter**.



The screenshot shows a terminal window titled "PLABSRV02" with a dark theme. The user is at the prompt "admin@localhost: /etc/yum.repos.d". They have entered the command "sudo dnf remove httpd". The terminal shows the password prompt, the command execution, and the output of the dnf command. The output lists the packages to be removed and their dependencies, along with a transaction summary.

```
admin@localhost: /etc/yum.repos.d
[admin@localhost yum.repos.d]$ sudo dnf remove httpd
[sudo] password for admin:
Dependencies resolved.
=====
Package                                Architecture Version                                Repository                                Size
=====
Removing:
httpd                                  x86_64      2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5 @appstream                                4.3 M
Removing unused dependencies:
almalinux-logos-httpd                 noarch      84.5-1.el8                                @appstream                                11 k
httpd-filesystem                       noarch      2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5 @appstream                                400 k
mod_http2                              x86_64      1.15.7-5.module_el8.6.0+2872+fe0ff7aa    @appstream                                394 k
=====
Transaction Summary
=====
Remove  4 Packages

Freed space: 4.7 M
Is this ok [y/N]: y
```


Figure 1.19 Screenshot of PLABSRV02: The Terminal window is displayed, showing the required command typed in and y entered for the Is this ok [y/N] prompt.

Step 2

The primary software package **httpd** is now removed.

```

PLABSRV02 Device Popup - Work - Microsoft Edge
https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABSRV02

PLABSRV02
Activities Terminal Menu May 9 12:04
admin@localhost:/etc/yum.repos.d

File Edit View Search Terminal Help
httpd-filesystem noarch 2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5 @appstream 400
mod_http2 x86_64 1.15.7-5.module_el8.6.0+2872+fe0ff7aa @appstream 394 k

Transaction Summary
=====
Remove 4 Packages

Freed space: 4.7 M
Is this ok [y/N]: y
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing : 1/1
  Running scriptlet: httpd-2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5.x86_64 1/1
  Running scriptlet: httpd-2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5.x86_64 1/4
  Erasing : httpd-2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5.x86_64 1/4
  Running scriptlet: httpd-2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5.x86_64 1/4
  Erasing : httpd-filesystem-2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5.noarch 2/4
  Running scriptlet: httpd-filesystem-2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5.noarch 2/4
  Erasing : almalinux-logos-httpd-84.5-1.el8.noarch 3/4
  Erasing : mod_http2-1.15.7-5.module_el8.6.0+2872+fe0ff7aa.x86_64 4/4
  Running scriptlet: mod_http2-1.15.7-5.module_el8.6.0+2872+fe0ff7aa.x86_64 4/4
  Verifying : almalinux-logos-httpd-84.5-1.el8.noarch 1/4
  Verifying : httpd-2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5.x86_64 2/4
  Verifying : httpd-filesystem-2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5.noarch 3/4
  Verifying : mod_http2-1.15.7-5.module_el8.6.0+2872+fe0ff7aa.x86_64 4/4

Removed:
  almalinux-logos-httpd-84.5-1.el8.noarch
  httpd-2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5.x86_64
  httpd-filesystem-2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5.noarch
  mod_http2-1.15.7-5.module_el8.6.0+2872+fe0ff7aa.x86_64

Complete!
[admin@localhost yum.repos.d]$

```

Figure 1.20 Screenshot of PLABSRV02: The Terminal window is displayed, showing the output of the executed command.

Step 3

In the **Terminal** window, type the following command and press **Enter**:

```
sudo dnf autoremove
```


Type the following **Password** and press **Enter**:

Passw0rd

Next, type the following for the **Is this ok [y/N]** prompt:

y

```

PLABSRV02 Device Popup - Work - Microsoft Edge
https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABSRV02

PLABSRV02
Activities Terminal Menu May 9 12:08
admin@localhost:/etc/yum.repos.d

File Edit View Search Terminal Help
Running scriptlet: httpd-2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5.x86_64 1/4
Erasing : httpd-filesystem-2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5.noarch 2/4
Running scriptlet: httpd-filesystem-2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5.noarch 2/4
Erasing : almalinux-logos-httpd-84.5-1.el8.noarch 3/4
Erasing : mod_http2-1.15.7-5.module_el8.6.0+2872+fe0ff7aa.x86_64 4/4
Running scriptlet: mod_http2-1.15.7-5.module_el8.6.0+2872+fe0ff7aa.x86_64 4/4
Verifying : almalinux-logos-httpd-84.5-1.el8.noarch 1/4
Verifying : httpd-2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5.x86_64 2/4
Verifying : httpd-filesystem-2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5.noarch 3/4
Verifying : mod_http2-1.15.7-5.module_el8.6.0+2872+fe0ff7aa.x86_64 4/4

Removed:
  almalinux-logos-httpd-84.5-1.el8.noarch
  httpd-2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5.x86_64
  httpd-filesystem-2.4.37-51.module_el8.7.0+3515+9e4fe0d6.5.noarch
  mod_http2-1.15.7-5.module_el8.6.0+2872+fe0ff7aa.x86_64

Complete!
[admin@localhost yum.repos.d]$ sudo dnf autoremove
[sudo] password for admin:
Last metadata expiration check: 0:15:35 ago on Tue 09 May 2023 11:52:26 AM BST.
Dependencies resolved.
=====
Package Architecture Version Repository Size
=====
Removing:
libibumad x86_64 41.0-1.el8 @baseos 41 k
python3-configobj noarch 5.0.6-11.el8 @anaconda 342 k
radvd x86_64 2.17-15.el8 @AppStream 188 k
rdma-core x86_64 41.0-1.el8 @baseos 103 k

Transaction Summary
=====
Remove 4 Packages

Freed space: 673 k
Is this ok [y/N]: y

```

Figure 1.21 Screenshot of PLABSRV02: The Terminal window is displayed, showing the required command entered and executed.

Step 4

The **dnf** command used with the **autoremove** parameter will remove all the orphaned packages from the previously removed software package, that is, from **httpd**.

```

admin@localhost:/etc/yum.repos.d
File Edit View Search Terminal Help
libibumad x86_64 41.0-1.el8 @baseos 41 k
python3-configobj noarch 5.0.6-11.el8 @anaconda 342 k
radvd x86_64 2.17-15.el8 @AppStream 188 k
rdma-core x86_64 41.0-1.el8 @baseos 103 k

Transaction Summary
=====
Remove 4 Packages

Freed space: 673 k
Is this ok [y/N]: y
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing : 1/1
  Running scriptlet: python3-configobj-5.0.6-11.el8.noarch 1/1
  Erasing : python3-configobj-5.0.6-11.el8.noarch 1/4
  Erasing : rdma-core-41.0-1.el8.x86_64 2/4
  Running scriptlet: rdma-core-41.0-1.el8.x86_64 2/4
  Running scriptlet: radvd-2.17-15.el8.x86_64 3/4
  Erasing : radvd-2.17-15.el8.x86_64 3/4
  Running scriptlet: radvd-2.17-15.el8.x86_64 3/4
  Erasing : libibumad-41.0-1.el8.x86_64 4/4
  Running scriptlet: libibumad-41.0-1.el8.x86_64 4/4
  Verifying : libibumad-41.0-1.el8.x86_64 1/4
  Verifying : python3-configobj-5.0.6-11.el8.noarch 2/4
  Verifying : radvd-2.17-15.el8.x86_64 3/4
  Verifying : rdma-core-41.0-1.el8.x86_64 4/4

Removed:
libibumad-41.0-1.el8.x86_64 python3-configobj-5.0.6-11.el8.noarch radvd-2.17-15.el8.x86_64
rdma-core-41.0-1.el8.x86_64

Complete!
[admin@localhost yum.repos.d]$

```

Figure 1.22 Screenshot of PLABSRV02: The Terminal window is displayed, showing the output of the command.

Note: The above orphaned packages were dependent on the **httpd** repository. Since we removed it, they are no longer referenced to the repository; hence can be removed to free up disk space.

Close the **Terminal** window.

Exercise 2 - Install and Update Software using Apt

Apt is a command line tool that works with Ubuntu's Advanced Packaging Tool (APT) to manage software packages.

In this exercise, you will use apt to install and update software packages.

Learning Outcomes

After completing this exercise, you should be able to:

- Use Apt to Explore and Install a Current Repository
- Install a Third-party Repository
- Upgrade and Remove Installed Software

Your Devices

You will be using the following device in this lab. Please power this on now.

- **PLABCLIENT01** - Ubuntu 20.04 - Stand-Alone Ubuntu Workstation (GUI)

Task 1 - Use Apt to Explore and Install a Current Repository

In this task, you will use apt to explore and install a current repository.

Step 1

Connect to **PLABCLIENT01**.

Login as **administrator** using the following **Password**:

Passw0rd

Press **Enter**.

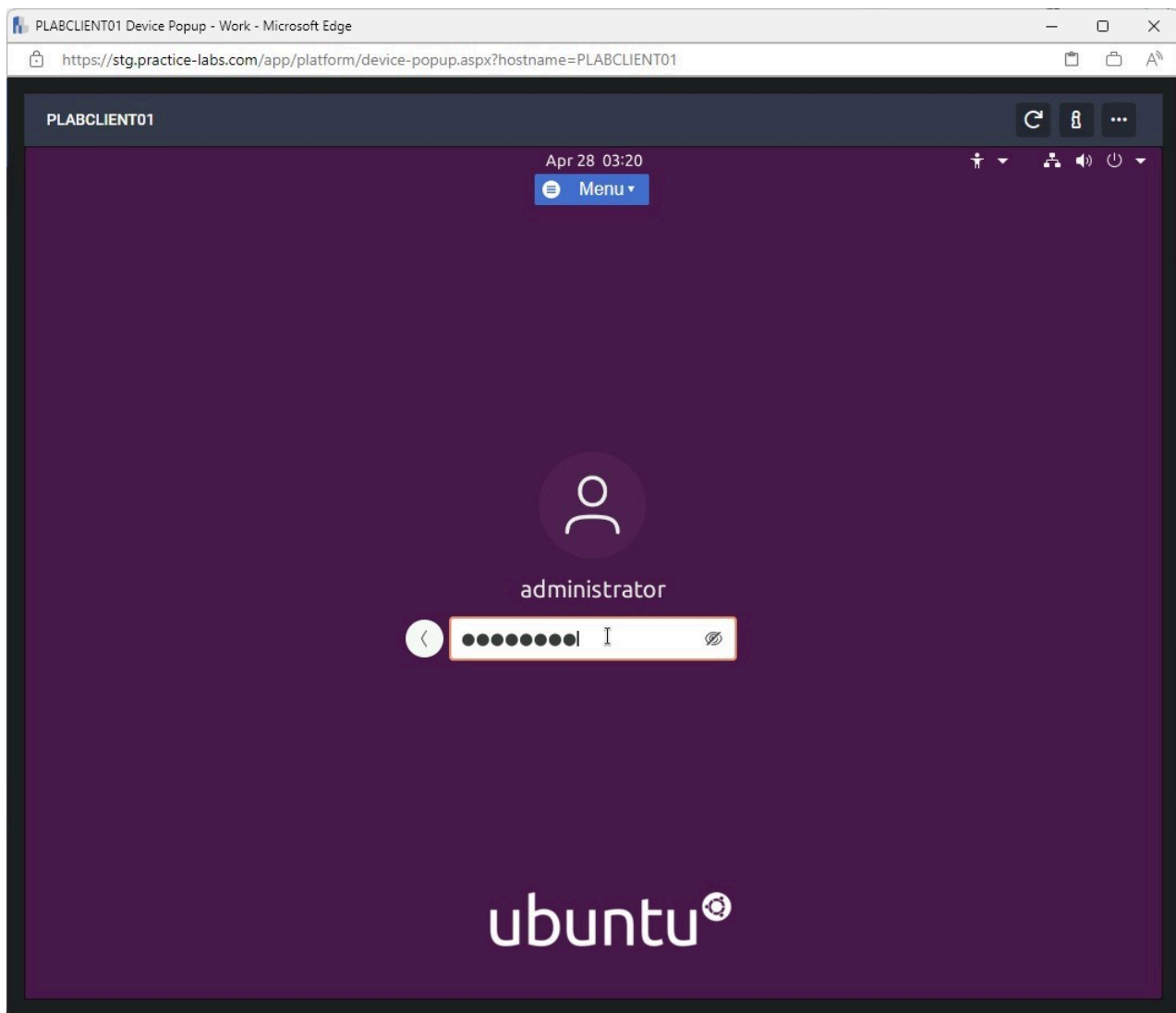


Figure 2.1 Screenshot of PLABCLIENT01: The login screen is displayed, showing the required credentials typed in.

Step 2

Click the **Show applications** icon on the bottom left corner.

Scroll down and select **Terminal**.

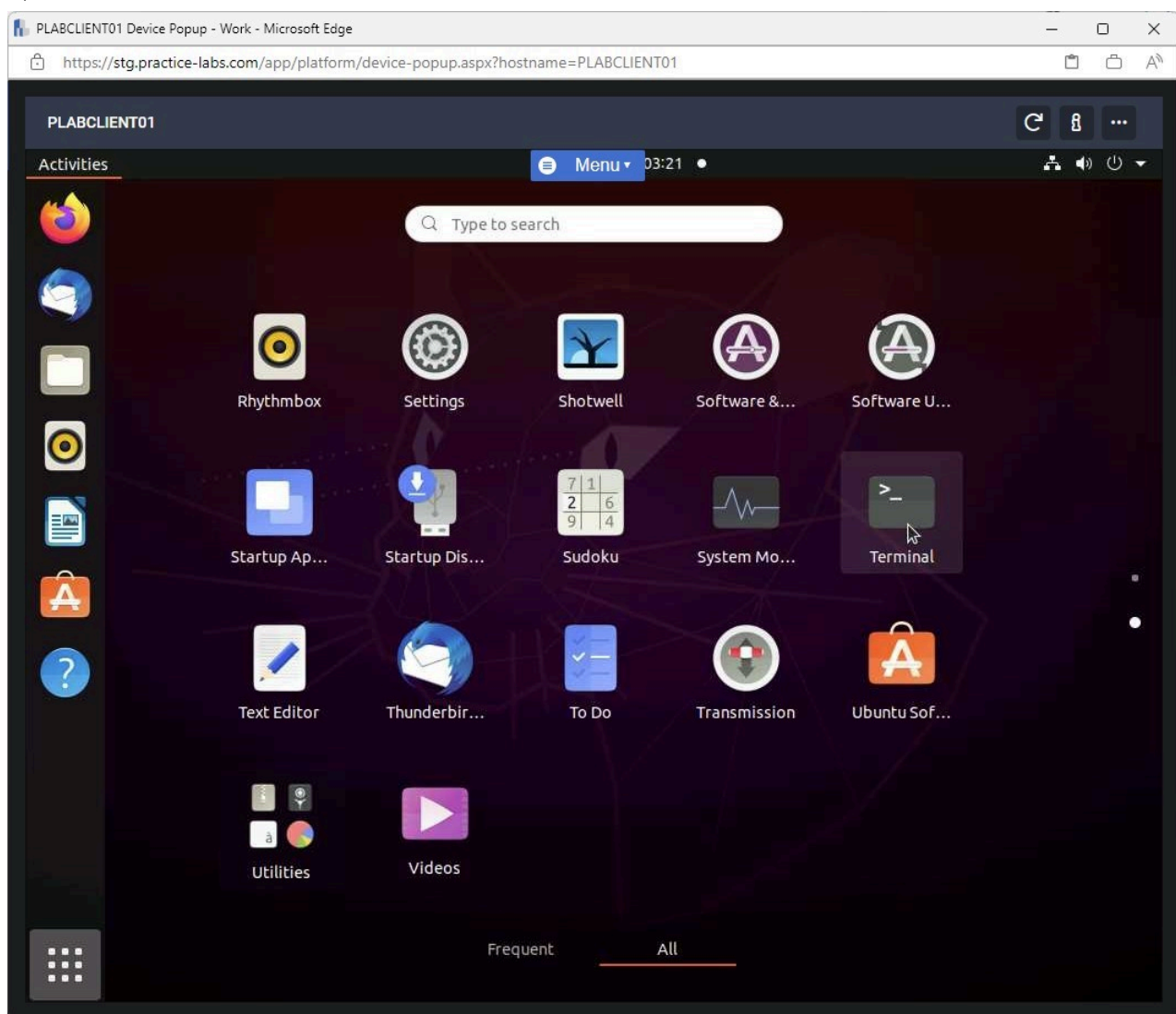


Figure 2.2 Screenshot of PLABCLIENT01: Show applications icon is clicked, and Terminal is selected.

Step 3

In the **Terminal** window, type the following command to navigate to the **etc/apt** directory:

```
cd /etc/apt
```

Press **Enter**.

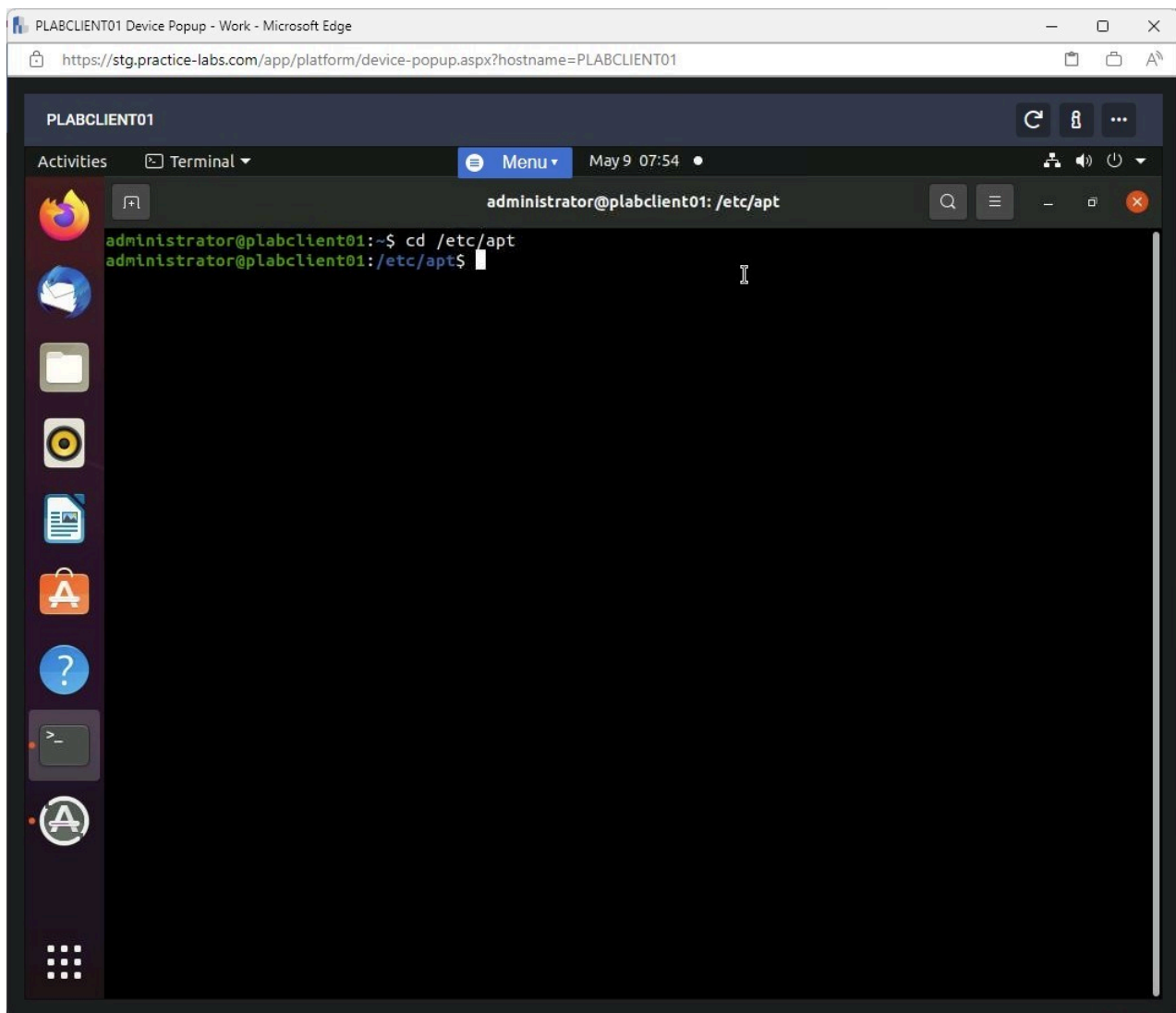


Figure 2.3 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Step 4

In the **Terminal** window, type the following command and press **Enter**:

```
ls -l
```

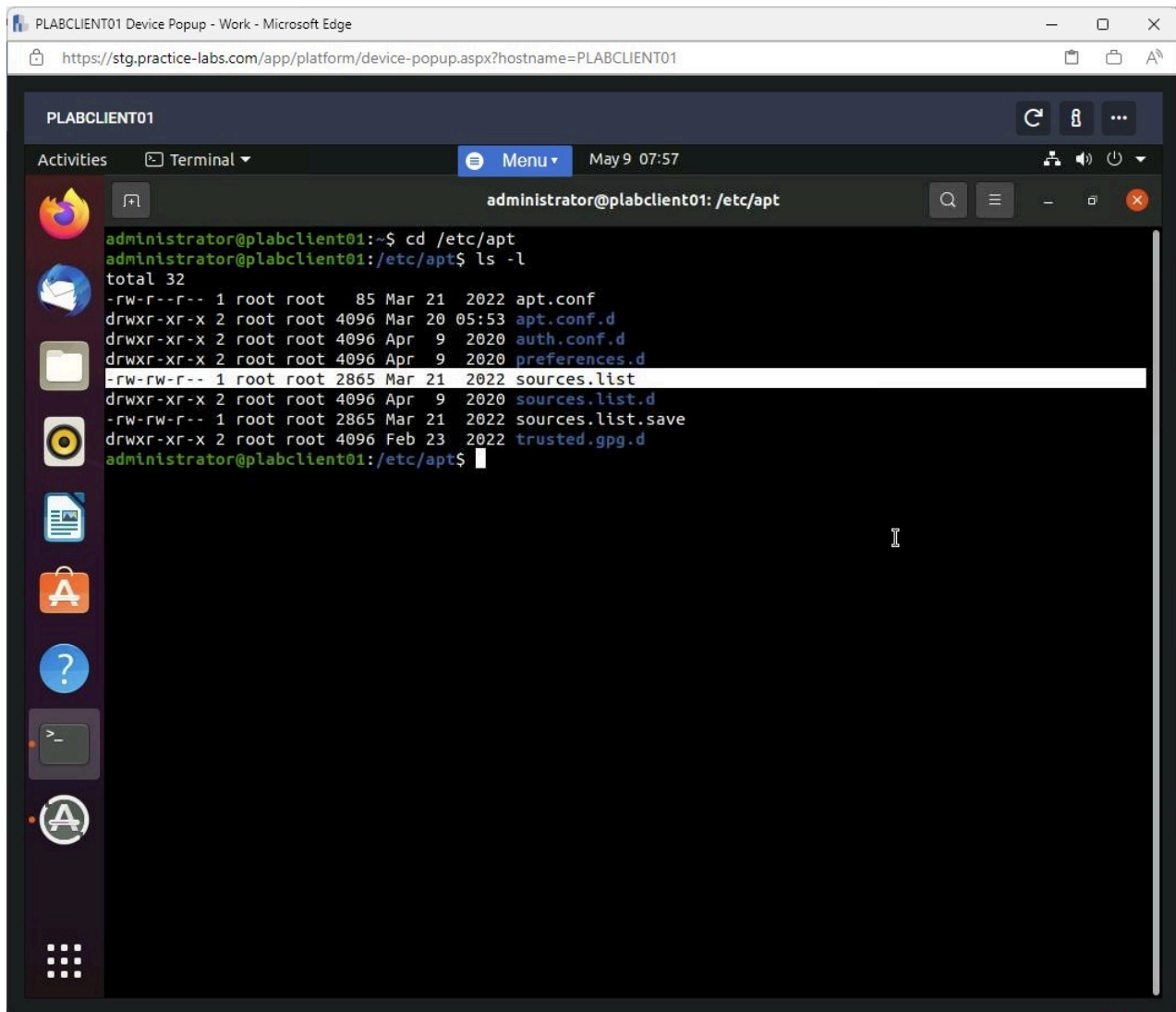



Figure 2.4 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: The ***sources.list*** file has a list of all the repositories that exist on the Linux system.

Step 5

In the **Terminal** window, type the following command and press **Enter**:

```
nano sources.list
```

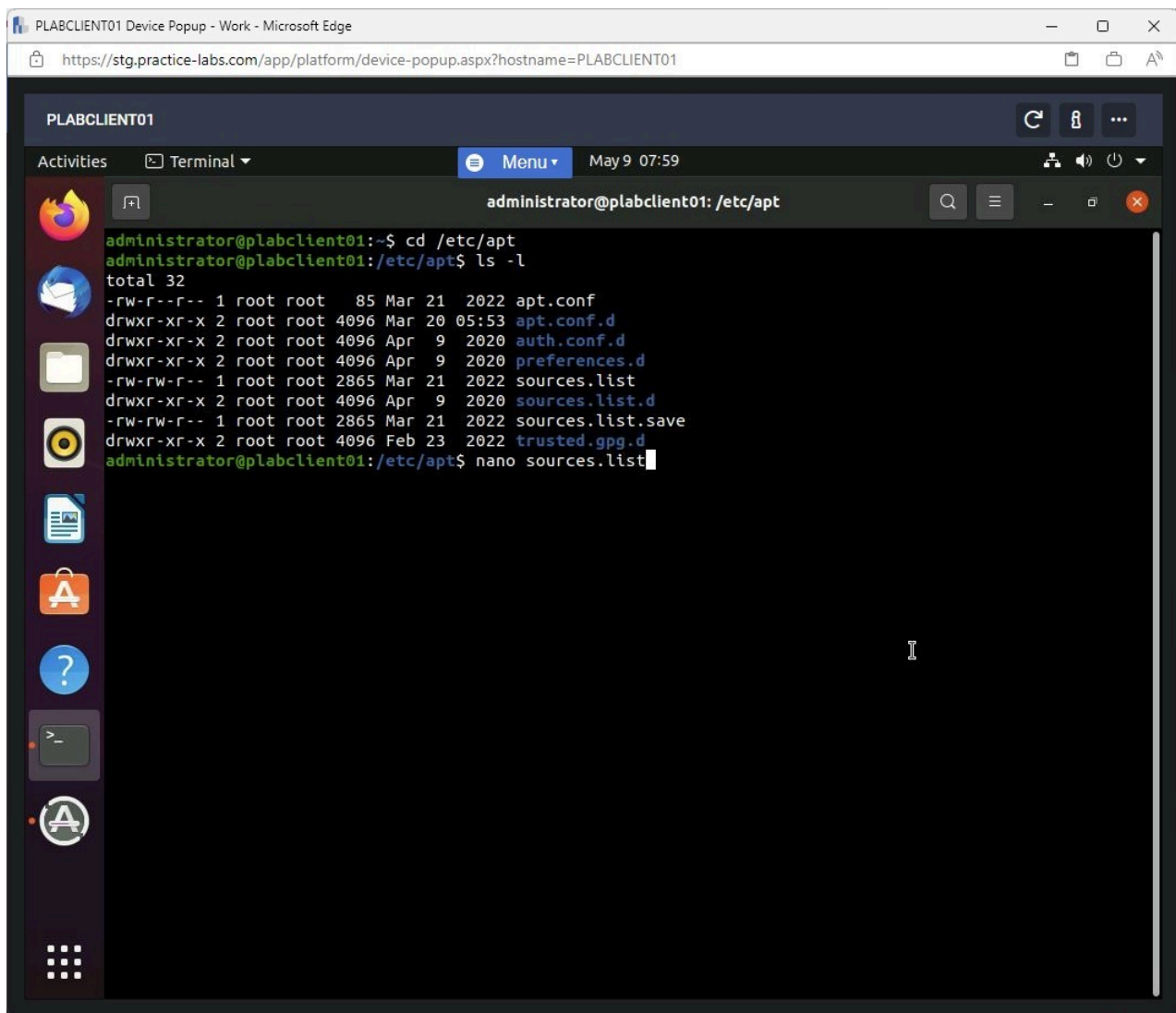


Figure 2.5 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered.

Step 6

The **sources.list** file is displayed.

```

PLABCLIENT01
Activities Terminal Menu May 9 08:02
administrator@plabclient01: /etc/apt
GNU nano 4.8 sources.list
deb cdrom:[Ubuntu 20.04.4 LTS _Focal Fossa_ - Release amd64 (20220223)]/ focal main restricted
# See http://help.ubuntu.com/community/UpgradeNotes for how to upgrade to
# newer versions of the distribution.
deb http://us.archive.ubuntu.com/ubuntu/ focal main restricted
# deb-src http://us.archive.ubuntu.com/ubuntu/ focal main restricted
## Major bug fix updates produced after the final release of the
## distribution.
# deb-src http://us.archive.ubuntu.com/ubuntu/ focal-updates main restricted
## N.B. software from this repository is ENTIRELY UNSUPPORTED by the Ubuntu
## team. Also, please note that software in universe WILL NOT receive any
## review or updates from the Ubuntu security team.
deb http://us.archive.ubuntu.com/ubuntu/ focal universe
# deb-src http://us.archive.ubuntu.com/ubuntu/ focal universe
# deb-src http://us.archive.ubuntu.com/ubuntu/ focal-updates universe
## N.B. software from this repository is ENTIRELY UNSUPPORTED by the Ubuntu
## team, and may not be under a free licence. Please satisfy yourself as to
## your rights to use the software. Also, please note that software in
## multiverse WILL NOT receive any review or updates from the Ubuntu
## security team.
deb http://us.archive.ubuntu.com/ubuntu/ focal multiverse
# deb-src http://us.archive.ubuntu.com/ubuntu/ focal multiverse
# deb-src http://us.archive.ubuntu.com/ubuntu/ focal-updates multiverse
## N.B. software from this repository may not have been tested as
## extensively as that contained in the main release, although it includes
## newer versions of some applications which may provide useful features.
## Also, please note that software in backports WILL NOT receive any review
## or updates from the Ubuntu security team.
# deb-src http://us.archive.ubuntu.com/ubuntu/ focal-backports main restricted universe multiverse
[ File 'sources.list' is unwritable ]
^G Get Help ^O Write Out ^W Where Is ^K Cut Text ^J Justify ^C Cur Pos M-U Undo
^X Exit ^R Read File ^\ Replace ^U Paste Text ^T To Spell ^_ Go To Line M-E Redo

```

Figure 2.6 Screenshot of PLABCLIENT01: The sources.list file is displayed.

Note: The **sources.list** file contains all the details on the repositories in the system. Each line that is not commented with a # is a repository. The following format is shown for a repository:

deb http://[Url of the repository] [distro] [type]

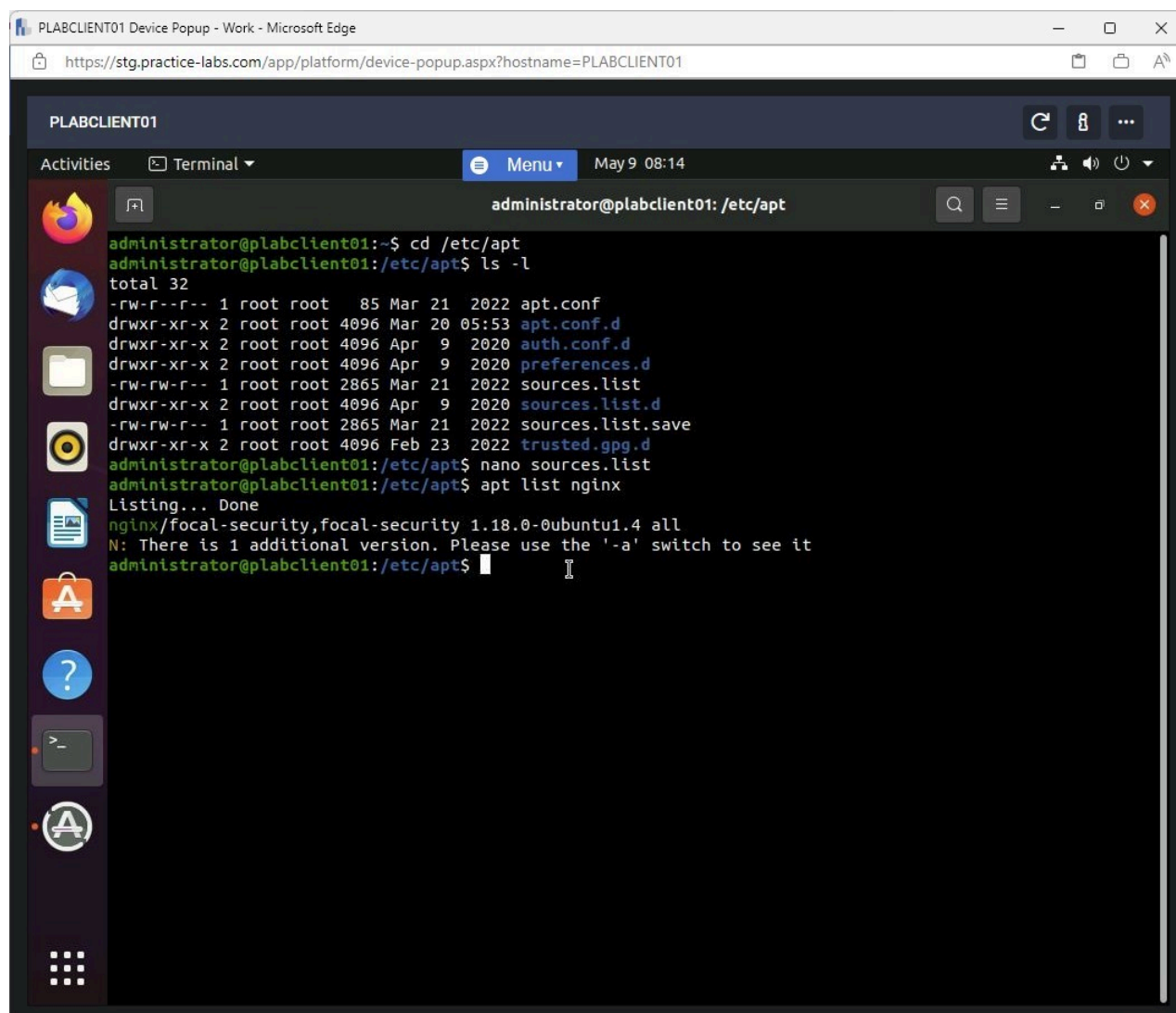
deb indicates the type of package used by the repository. The second field is the URL of the repository, the third field is the Ubuntu installation version, and the final field indicates the type of repository.

Step 7

Press the **Ctrl+X** keys on the keyboard to exit the file.

Back on the **Terminal** window, type the following command and press **Enter**:

```
apt list nginx
```



The screenshot shows a terminal window titled 'PLABCLIENT01' with a dark background. The prompt is 'administrator@plabclient01: /etc/apt'. The user has entered 'cd /etc/apt' and 'ls -l', which lists files in /etc/apt including apt.conf, apt.conf.d, auth.conf.d, preferences.d, sources.list, sources.list.d, sources.list.save, and trusted.gpg.d. Then, the user enters 'nano sources.list' and 'apt list nginx'. The output shows 'nginx/focal-security,focal-security 1.18.0-0ubuntu1.4 all' and a message about an additional version. The terminal window is part of a desktop environment with a sidebar on the left containing icons for Firefox, Mail, Files, Music, Videos, Applications, and a search icon. The top bar shows 'May 9 08:14' and system icons.

```
administrator@plabclient01:~$ cd /etc/apt
administrator@plabclient01:/etc/apt$ ls -l
total 32
-rw-r--r-- 1 root root 85 Mar 21 2022 apt.conf
drwxr-xr-x 2 root root 4096 Mar 20 05:53 apt.conf.d
drwxr-xr-x 2 root root 4096 Apr 9 2020 auth.conf.d
drwxr-xr-x 2 root root 4096 Apr 9 2020 preferences.d
-rw-rw-r-- 1 root root 2865 Mar 21 2022 sources.list
drwxr-xr-x 2 root root 4096 Apr 9 2020 sources.list.d
-rw-rw-r-- 1 root root 2865 Mar 21 2022 sources.list.save
drwxr-xr-x 2 root root 4096 Feb 23 2022 trusted.gpg.d
administrator@plabclient01:/etc/apt$ nano sources.list
administrator@plabclient01:/etc/apt$ apt list nginx
Listing... Done
nginx/focal-security,focal-security 1.18.0-0ubuntu1.4 all
N: There is 1 additional version. Please use the '-a' switch to see it
administrator@plabclient01:/etc/apt$
```

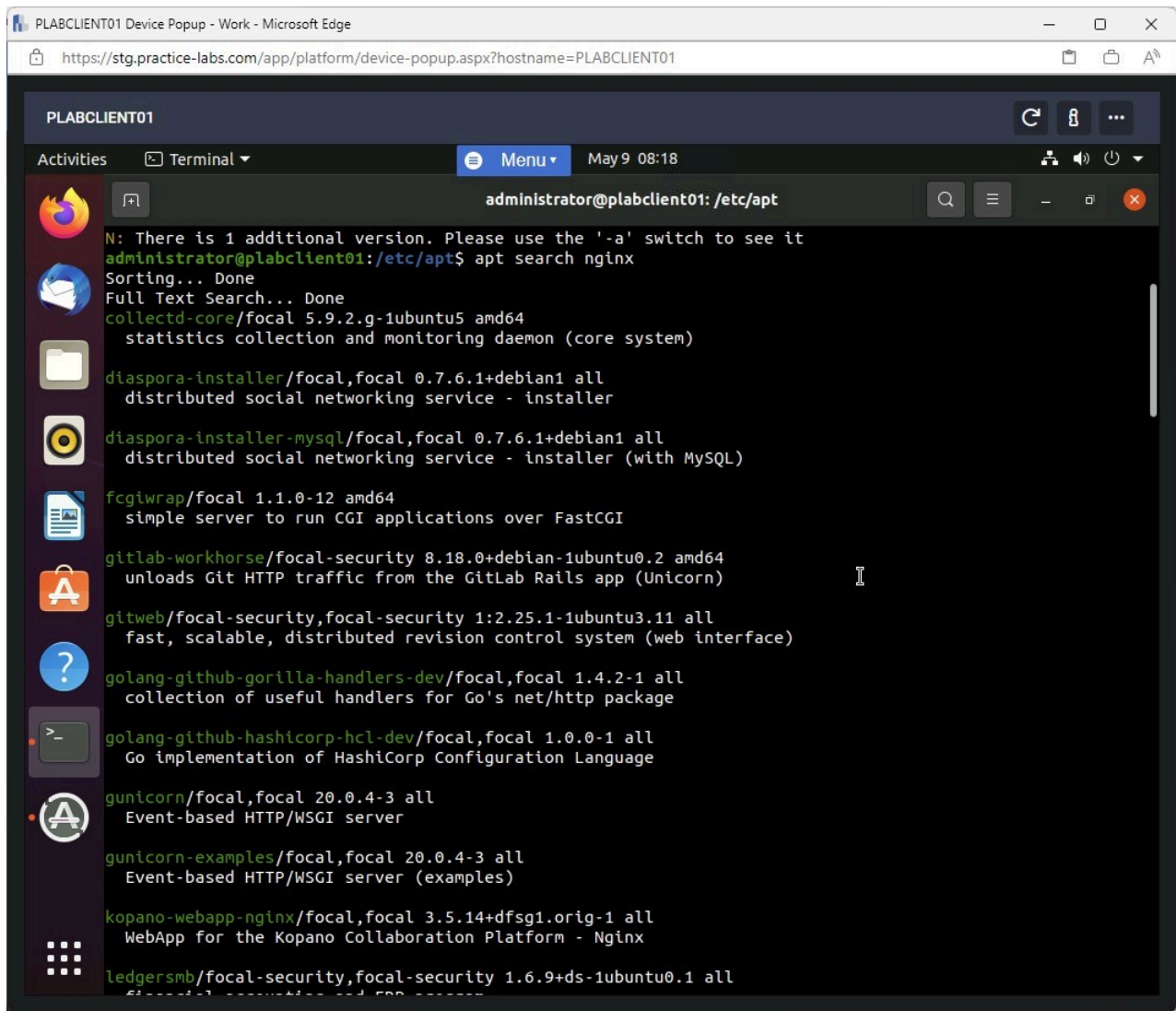
Figure 2.7 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: The **apt list nginx** command will list the **nginx** software package that is available.

Step 8

In the **Terminal** window, type the following command and press **Enter**:

```
apt search nginx
```



The screenshot shows a terminal window titled 'administrator@plabclient01: /etc/apt'. The command 'apt search nginx' has been executed, resulting in a list of software packages. The output includes:

```
N: There is 1 additional version. Please use the '-a' switch to see it
administrator@plabclient01:/etc/apt$ apt search nginx
Sorting... Done
Full Text Search... Done
collectd-core/focal 5.9.2.g-1ubuntu5 amd64
  statistics collection and monitoring daemon (core system)

diaspora-installer/focal,focal 0.7.6.1+debian1 all
  distributed social networking service - installer

diaspora-installer-mysql/focal,focal 0.7.6.1+debian1 all
  distributed social networking service - installer (with MySQL)

fcgiwrap/focal 1.1.0-12 amd64
  simple server to run CGI applications over FastCGI

gitlab-workhorse/focal-security 8.18.0+debian-1ubuntu0.2 amd64
  unloads Git HTTP traffic from the GitLab Rails app (Unicorn)

gitweb/focal-security,focal-security 1:2.25.1-1ubuntu3.11 all
  fast, scalable, distributed revision control system (web interface)

golang-github-gorilla-handlers-dev/focal,focal 1.4.2-1 all
  collection of useful handlers for Go's net/http package

golang-github-hashicorp-hcl-dev/focal,focal 1.0.0-1 all
  Go implementation of HashiCorp Configuration Language

gunicorn/focal,focal 20.0.4-3 all
  Event-based HTTP/WSGI server

gunicorn-examples/focal,focal 20.0.4-3 all
  Event-based HTTP/WSGI server (examples)

kopano-webapp-nginx/focal,focal 3.5.14+dfsg1.orig-1 all
  WebApp for the Kopano Collaboration Platform - Nginx

ledgersmb/focal-security,focal-security 1.6.9+ds-1ubuntu0.1 all
  Samba client for LedgerSMB
```

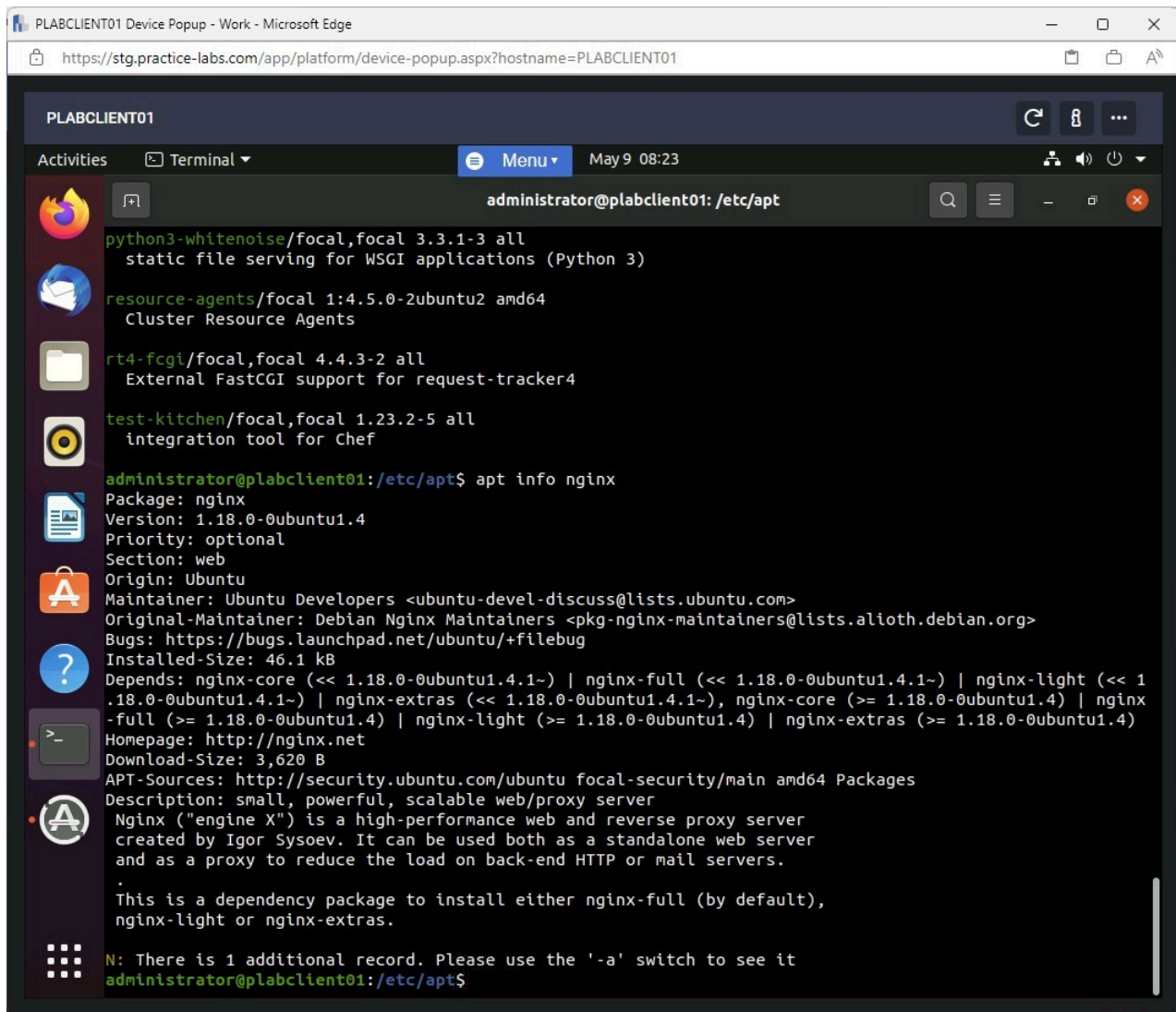
Figure 2.8 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: The *search* parameter conducts a complete search for the nginx software package.

Step 9

In the **Terminal** window, type the following command and press **Enter**:

```
apt info nginx
```

The screenshot shows a terminal window titled 'PLABCLIENT01' with the prompt 'administrator@plabclient01: /etc/apt'. The terminal displays the output of the command 'apt info nginx'. The output lists several installed packages: 'python3-whitenoise', 'resource-agents', 'rt4-fcgi', and 'test-kitchen'. It then shows the details for the 'nginx' package, including its version (1.18.0-0ubuntu1.4), priority (optional), section (web), origin (Ubuntu), maintainer (Ubuntu Developers), original maintainer (Debian Nginx Maintainers), bugs (https://bugs.launchpad.net/ubuntu/+filebug), installed size (46.1 kB), dependencies (nginx-core, nginx-full, nginx-light, nginx-extras, nginx-core), homepage (http://nginx.net), download size (3,620 B), and APT sources (http://security.ubuntu.com/ubuntu focal-security/main amd64 Packages). The description states that Nginx is a high-performance web and reverse proxy server created by Igor Sysoev. It can be used both as a standalone web server and as a proxy to reduce the load on back-end HTTP or mail servers. The terminal also shows a note that there is 1 additional record for nginx and suggests using the '-a' switch to see it.

```
python3-whitenoise/focal,focal 3.3.1-3 all
static file serving for WSGI applications (Python 3)

resource-agents/focal 1:4.5.0-2ubuntu2 amd64
Cluster Resource Agents

rt4-fcgi/focal,focal 4.4.3-2 all
External FastCGI support for request-tracker4

test-kitchen/focal,focal 1.23.2-5 all
integration tool for Chef

administrator@plabclient01:/etc/apt$ apt info nginx
Package: nginx
Version: 1.18.0-0ubuntu1.4
Priority: optional
Section: web
Origin: Ubuntu
Maintainer: Ubuntu Developers <ubuntu-devel-discuss@lists.ubuntu.com>
Original-Maintainer: Debian Nginx Maintainers <pkg-nginx-maintainers@lists.aliases.debian.org>
Bugs: https://bugs.launchpad.net/ubuntu/+filebug
Installed-Size: 46.1 kB
Depends: nginx-core (<< 1.18.0-0ubuntu1.4.1~) | nginx-full (<< 1.18.0-0ubuntu1.4.1~) | nginx-light (<< 1.18.0-0ubuntu1.4.1~) | nginx-extras (<< 1.18.0-0ubuntu1.4.1~), nginx-core (>= 1.18.0-0ubuntu1.4) | nginx-full (>= 1.18.0-0ubuntu1.4) | nginx-light (>= 1.18.0-0ubuntu1.4) | nginx-extras (>= 1.18.0-0ubuntu1.4)
Homepage: http://nginx.net
Download-Size: 3,620 B
APT-Sources: http://security.ubuntu.com/ubuntu focal-security/main amd64 Packages
Description: small, powerful, scalable web/proxy server
 Nginx ("engine X") is a high-performance web and reverse proxy server
 created by Igor Sysoev. It can be used both as a standalone web server
 and as a proxy to reduce the load on back-end HTTP or mail servers.
.
 This is a dependency package to install either nginx-full (by default),
 nginx-light or nginx-extras.

N: There is 1 additional record. Please use the '-a' switch to see it
administrator@plabclient01:/etc/apt$
```

Figure 2.9 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

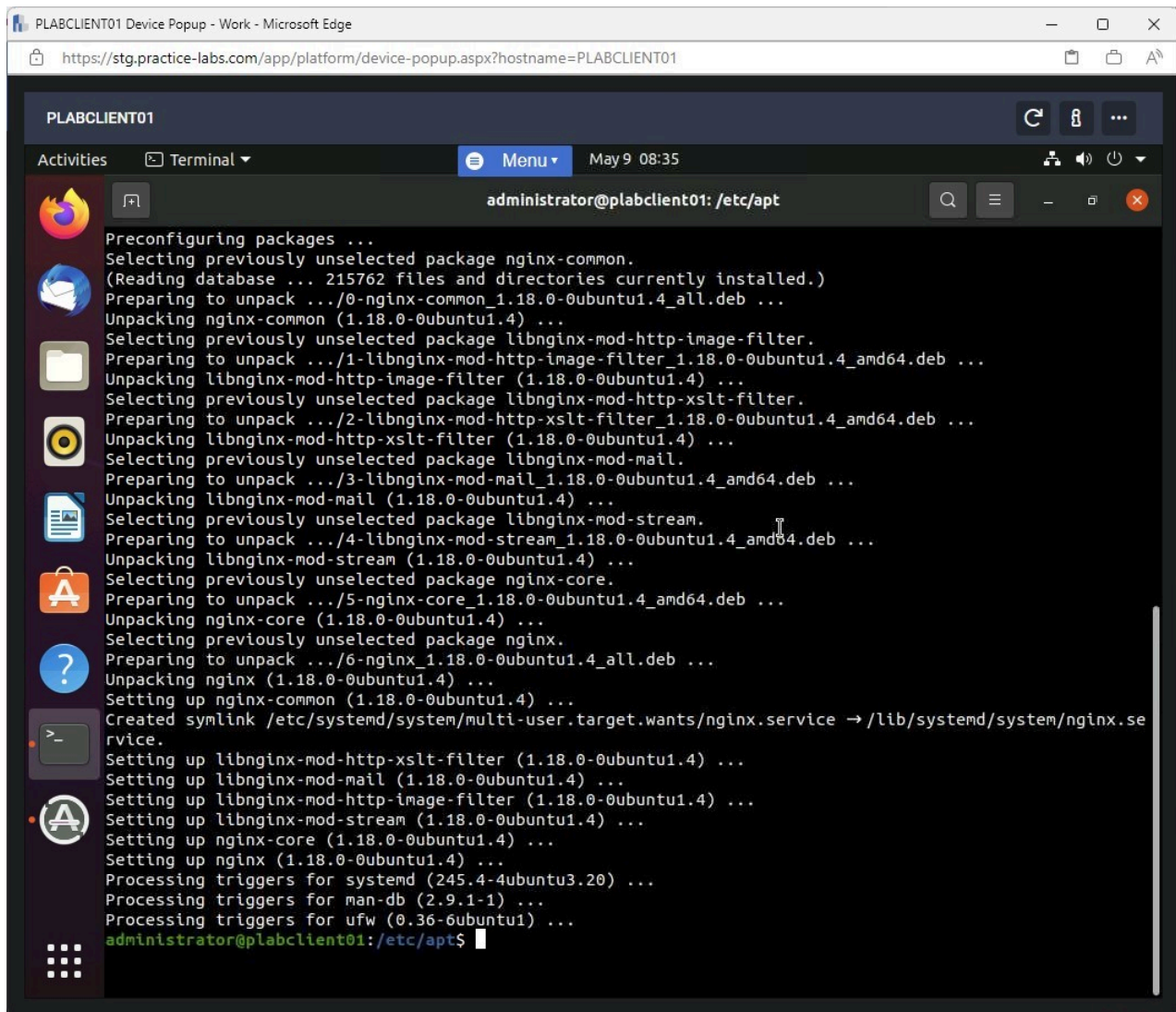
Note: The *info* parameter will list all information on the **nginx** package.

Step 10

In the **Terminal** window, type the following command and press **Enter**:

```
sudo apt install nginx
```

Type the following **Password** and press **Enter**:



```
PLABCLIENT01 Device Popup - Work - Microsoft Edge
https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABCLIENT01

PLABCLIENT01
Activities Terminal Menu May 9 08:35
administrator@plabclient01: /etc/apt

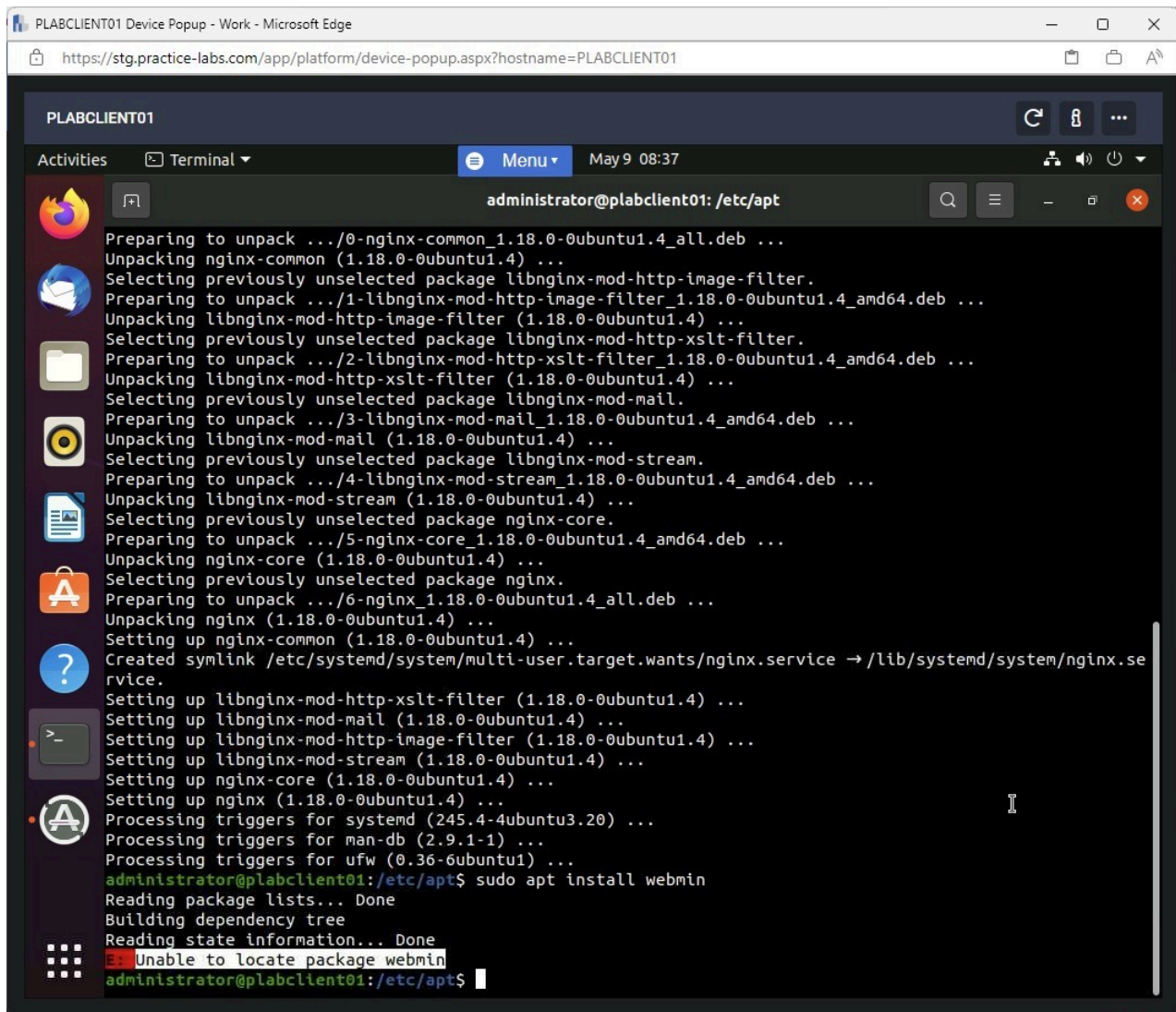
Preconfiguring packages ...
Selecting previously unselected package nginx-common.
(Reading database ... 215762 files and directories currently installed.)
Preparing to unpack .../0-nginx-common_1.18.0-0ubuntu1.4_all.deb ...
Unpacking nginx-common (1.18.0-0ubuntu1.4) ...
Selecting previously unselected package libnginx-mod-http-image-filter.
Preparing to unpack .../1-libnginx-mod-http-image-filter_1.18.0-0ubuntu1.4_amd64.deb ...
Unpacking libnginx-mod-http-image-filter (1.18.0-0ubuntu1.4) ...
Selecting previously unselected package libnginx-mod-http-xslt-filter.
Preparing to unpack .../2-libnginx-mod-http-xslt-filter_1.18.0-0ubuntu1.4_amd64.deb ...
Unpacking libnginx-mod-http-xslt-filter (1.18.0-0ubuntu1.4) ...
Selecting previously unselected package libnginx-mod-mail.
Preparing to unpack .../3-libnginx-mod-mail_1.18.0-0ubuntu1.4_amd64.deb ...
Unpacking libnginx-mod-mail (1.18.0-0ubuntu1.4) ...
Selecting previously unselected package libnginx-mod-stream.
Preparing to unpack .../4-libnginx-mod-stream_1.18.0-0ubuntu1.4_amd64.deb ...
Unpacking libnginx-mod-stream (1.18.0-0ubuntu1.4) ...
Selecting previously unselected package nginx-core.
Preparing to unpack .../5-nginx-core_1.18.0-0ubuntu1.4_amd64.deb ...
Unpacking nginx-core (1.18.0-0ubuntu1.4) ...
Selecting previously unselected package nginx.
Preparing to unpack .../6-nginx_1.18.0-0ubuntu1.4_all.deb ...
Unpacking nginx (1.18.0-0ubuntu1.4) ...
Setting up nginx-common (1.18.0-0ubuntu1.4) ...
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /lib/systemd/system/nginx.service.
Setting up libnginx-mod-http-xslt-filter (1.18.0-0ubuntu1.4) ...
Setting up libnginx-mod-mail (1.18.0-0ubuntu1.4) ...
Setting up libnginx-mod-http-image-filter (1.18.0-0ubuntu1.4) ...
Setting up libnginx-mod-stream (1.18.0-0ubuntu1.4) ...
Setting up nginx-core (1.18.0-0ubuntu1.4) ...
Setting up nginx (1.18.0-0ubuntu1.4) ...
Processing triggers for systemd (245.4-4ubuntu3.20) ...
Processing triggers for man-db (2.9.1-1) ...
Processing triggers for ufw (0.36-6ubuntu1) ...
administrator@plabclient01: /etc/apt$
```

Figure 2.11 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the output of the command.

Step 12

In the **Terminal** window, type the following command and press **Enter**:

```
sudo apt install webmin
```



```
PLABCLIENT01 Device Popup - Work - Microsoft Edge
https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABCLIENT01

PLABCLIENT01
Activities Terminal May 9 08:37
administrator@plabclient01: /etc/apt

Preparing to unpack .../0-nginx-common_1.18.0-0ubuntu1.4_all.deb ...
Unpacking nginx-common (1.18.0-0ubuntu1.4) ...
Selecting previously unselected package libnginx-mod-http-image-filter.
Preparing to unpack .../1-libnginx-mod-http-image-filter_1.18.0-0ubuntu1.4_amd64.deb ...
Unpacking libnginx-mod-http-image-filter (1.18.0-0ubuntu1.4) ...
Selecting previously unselected package libnginx-mod-http-xslt-filter.
Preparing to unpack .../2-libnginx-mod-http-xslt-filter_1.18.0-0ubuntu1.4_amd64.deb ...
Unpacking libnginx-mod-http-xslt-filter (1.18.0-0ubuntu1.4) ...
Selecting previously unselected package libnginx-mod-mail.
Preparing to unpack .../3-libnginx-mod-mail_1.18.0-0ubuntu1.4_amd64.deb ...
Unpacking libnginx-mod-mail (1.18.0-0ubuntu1.4) ...
Selecting previously unselected package libnginx-mod-stream.
Preparing to unpack .../4-libnginx-mod-stream_1.18.0-0ubuntu1.4_amd64.deb ...
Unpacking libnginx-mod-stream (1.18.0-0ubuntu1.4) ...
Selecting previously unselected package nginx-core.
Preparing to unpack .../5-nginx-core_1.18.0-0ubuntu1.4_amd64.deb ...
Unpacking nginx-core (1.18.0-0ubuntu1.4) ...
Selecting previously unselected package nginx.
Preparing to unpack .../6-nginx_1.18.0-0ubuntu1.4_all.deb ...
Unpacking nginx (1.18.0-0ubuntu1.4) ...
Setting up nginx-common (1.18.0-0ubuntu1.4) ...
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /lib/systemd/system/nginx.service.
Setting up libnginx-mod-http-xslt-filter (1.18.0-0ubuntu1.4) ...
Setting up libnginx-mod-mail (1.18.0-0ubuntu1.4) ...
Setting up libnginx-mod-http-image-filter (1.18.0-0ubuntu1.4) ...
Setting up libnginx-mod-stream (1.18.0-0ubuntu1.4) ...
Setting up nginx-core (1.18.0-0ubuntu1.4) ...
Setting up nginx (1.18.0-0ubuntu1.4) ...
Processing triggers for systemd (245.4-4ubuntu3.20) ...
Processing triggers for man-db (2.9.1-1) ...
Processing triggers for ufw (0.36-6ubuntu1) ...
administrator@plabclient01:/etc/apt$ sudo apt install webmin
Reading package lists... Done
Building dependency tree
Reading state information... Done
E: Unable to locate package webmin
administrator@plabclient01:/etc/apt$
```

Figure 2.12 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: Notice no matches were found for the **webmin** software package.

Keep the **Terminal** window open.

Task 2 - Install a Third-party Repository

In this task, you will install a third-party repository.

Step 1

Ensure you are connected to **PLABCLIENT01** and the **Terminal** window is open.

In the **Terminal** window, type the following command and press **Enter**:


```
sudo add-apt-repository "deb
http://download.webmin.com/download/repository sarge
contrib"
```

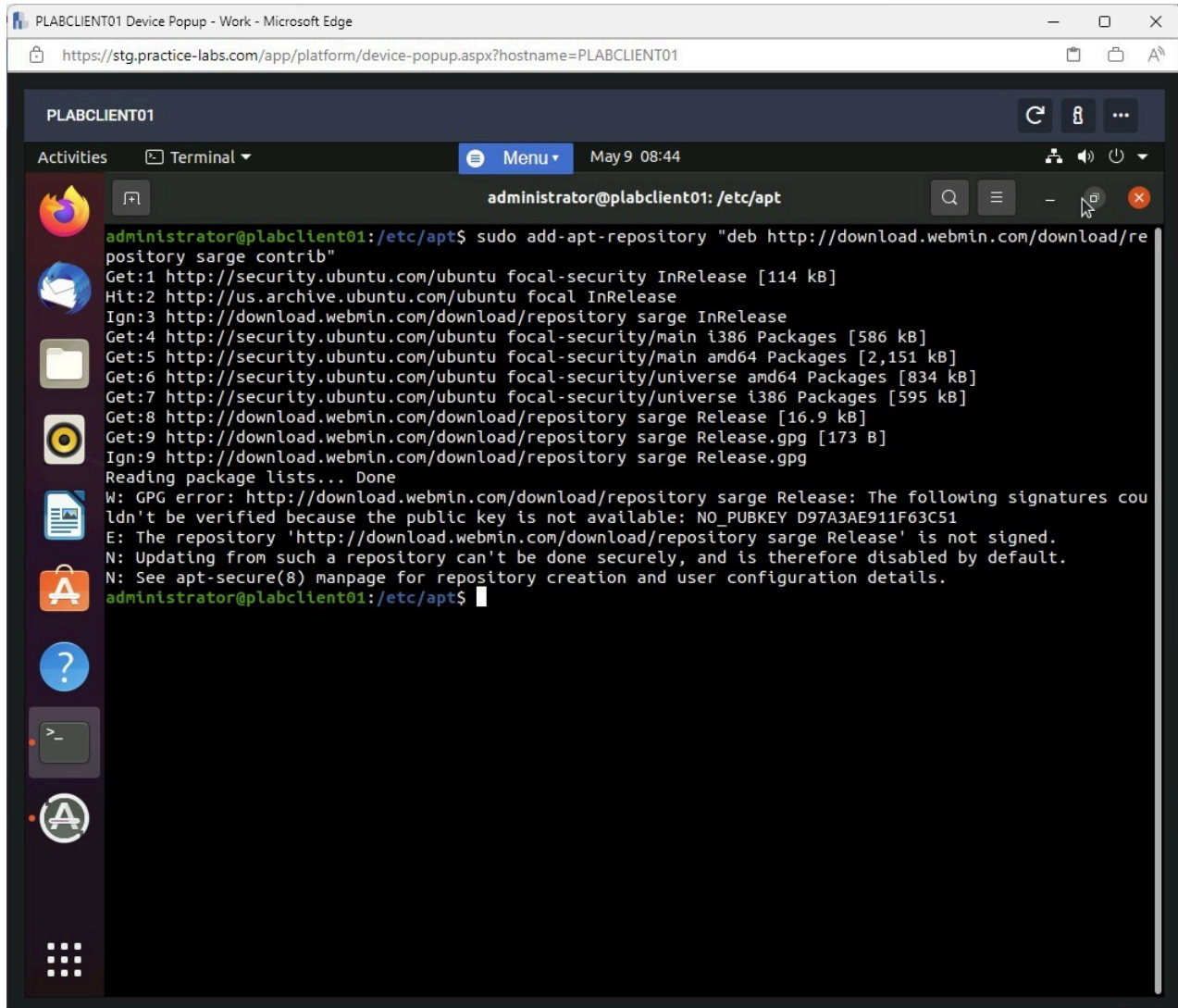


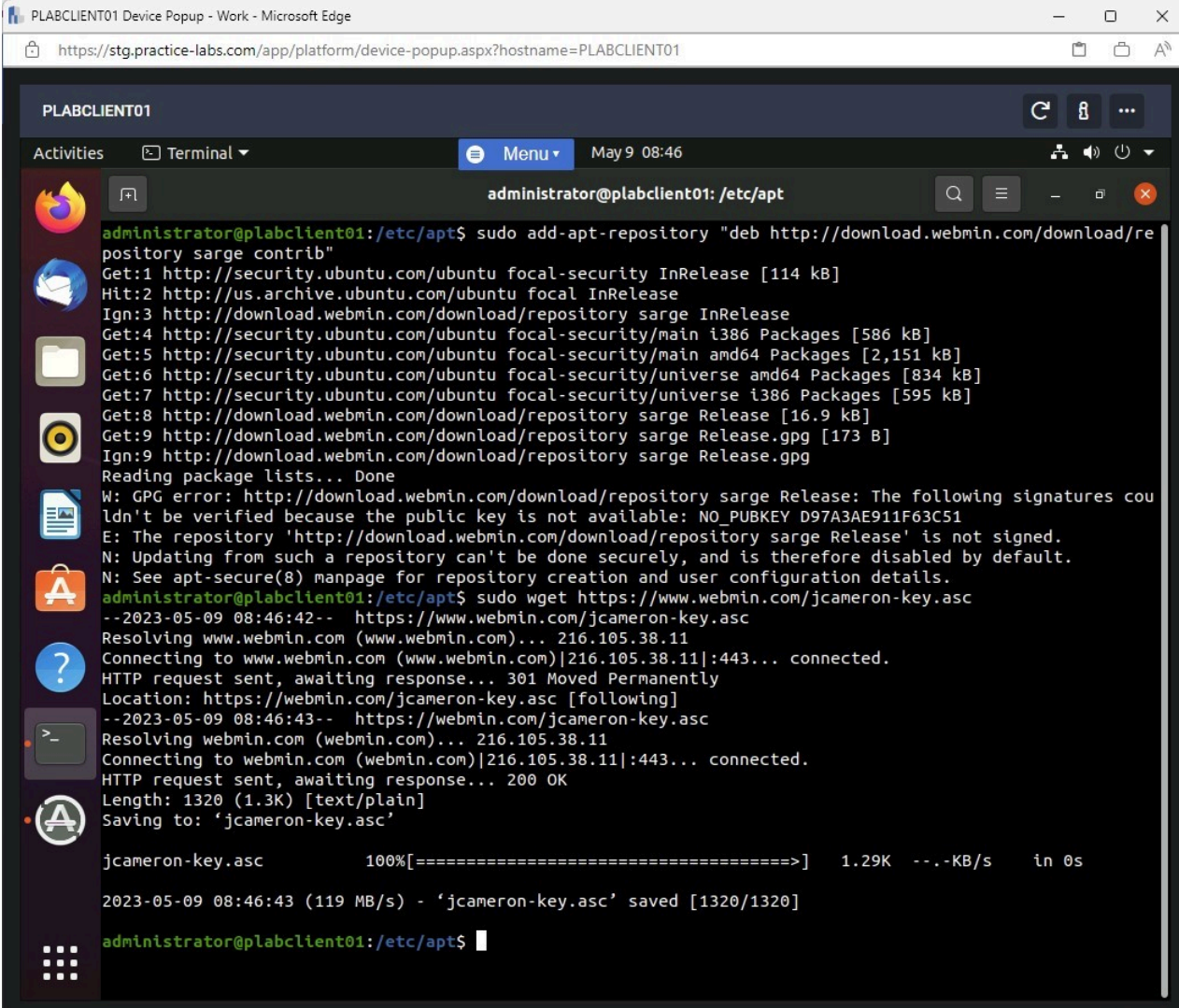
Figure 2.13 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Step 2

Next, you will import the signing key.

In the **Terminal** window, type the following command and press **Enter**:

```
sudo wget https://www.webmin.com/jcameron-key.asc
```



The screenshot shows a terminal window titled 'PLABCLIENT01' with the prompt 'administrator@plabclient01: /etc/apt'. The terminal output shows the execution of the command 'sudo add-apt-repository "deb http://download.webmin.com/download/repository sarge contrib"'. This command adds a new repository. The terminal then shows the execution of 'sudo wget https://www.webmin.com/jcameron-key.asc', which downloads a GPG key from the webmin website. The output of the wget command shows the file being saved as 'jcameron-key.asc'.

```
administrator@plabclient01: /etc/apt$ sudo add-apt-repository "deb http://download.webmin.com/download/repository sarge contrib"
Get:1 http://security.ubuntu.com/ubuntu focal-security InRelease [114 kB]
Hit:2 http://us.archive.ubuntu.com/ubuntu focal InRelease
Ign:3 http://download.webmin.com/download/repository sarge InRelease
Get:4 http://security.ubuntu.com/ubuntu focal-security/main i386 Packages [586 kB]
Get:5 http://security.ubuntu.com/ubuntu focal-security/main amd64 Packages [2,151 kB]
Get:6 http://security.ubuntu.com/ubuntu focal-security/universe amd64 Packages [834 kB]
Get:7 http://security.ubuntu.com/ubuntu focal-security/universe i386 Packages [595 kB]
Get:8 http://download.webmin.com/download/repository sarge Release [16.9 kB]
Get:9 http://download.webmin.com/download/repository sarge Release.gpg [173 B]
Ign:9 http://download.webmin.com/download/repository sarge Release.gpg
Reading package lists... Done
W: GPG error: http://download.webmin.com/download/repository sarge Release: The following signatures could not be verified because the public key is not available: NO_PUBKEY D97A3AE911F63C51
E: The repository 'http://download.webmin.com/download/repository sarge Release' is not signed.
N: Updating from such a repository can't be done securely, and is therefore disabled by default.
N: See apt-get(8) manpage for repository creation and user configuration details.
administrator@plabclient01: /etc/apt$ sudo wget https://www.webmin.com/jcameron-key.asc
--2023-05-09 08:46:42-- https://www.webmin.com/jcameron-key.asc
Resolving www.webmin.com (www.webmin.com)... 216.105.38.11
Connecting to www.webmin.com (www.webmin.com)|216.105.38.11|:443... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://webmin.com/jcameron-key.asc [following]
--2023-05-09 08:46:43-- https://webmin.com/jcameron-key.asc
Resolving webmin.com (webmin.com)... 216.105.38.11
Connecting to webmin.com (webmin.com)|216.105.38.11|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1320 (1.3K) [text/plain]
Saving to: 'jcameron-key.asc'

jcameron-key.asc      100%[=====] 1.29K  --.-KB/s  in 0s

2023-05-09 08:46:43 (119 MB/s) - 'jcameron-key.asc' saved [1320/1320]

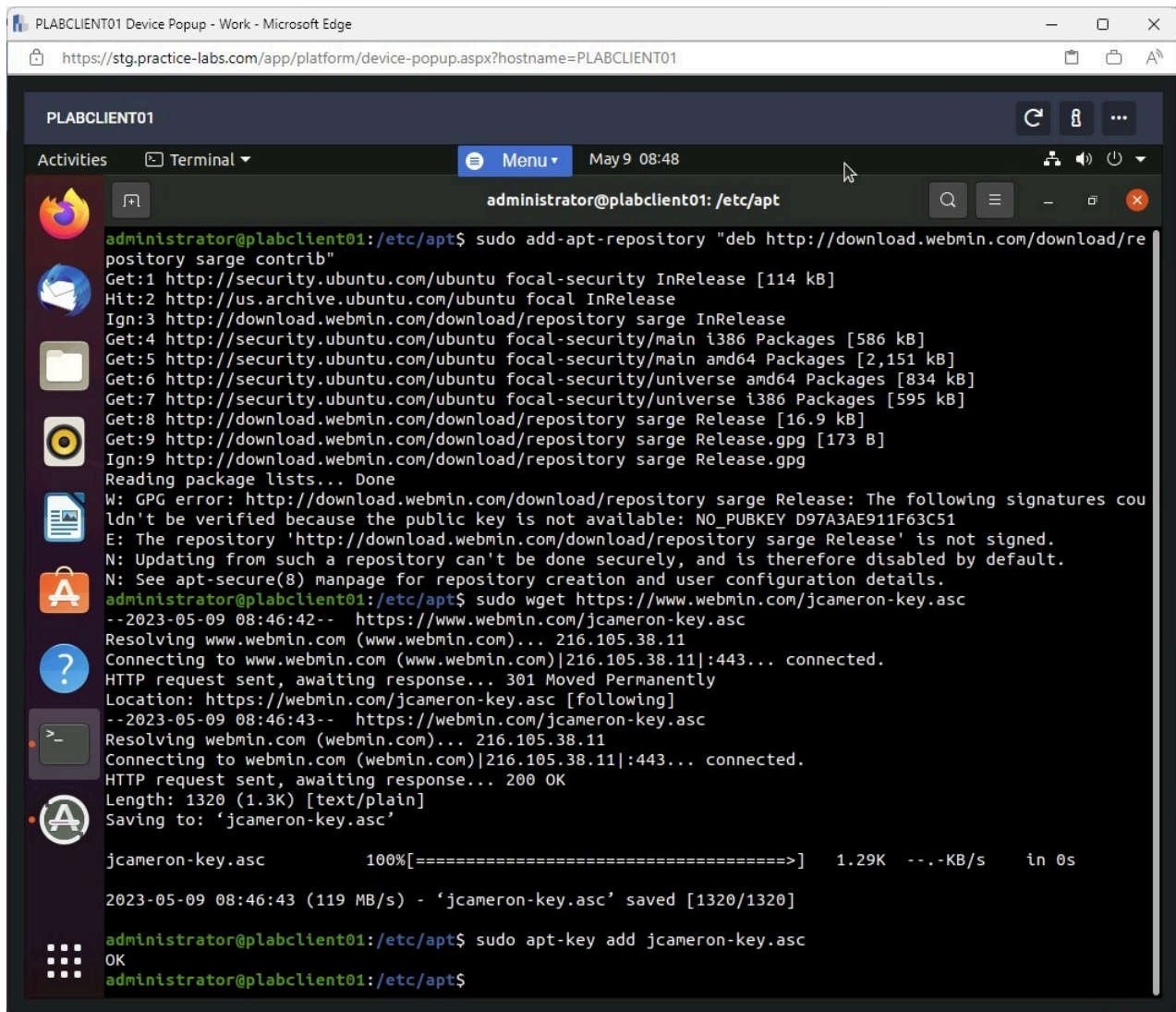
administrator@plabclient01: /etc/apt$
```

Figure 2.14 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Step 3

In the **Terminal** window, type the following command and press **Enter**:

```
sudo apt-key add jcameron-key.asc
```



The screenshot shows a terminal window titled 'PLABCLIENT01' with the prompt 'administrator@plabclient01: /etc/apt'. The terminal output shows the execution of the command 'sudo add-apt-repository "deb http://download.webmin.com/download/repository sarge contrib"'. This command adds a new repository. The terminal then shows the execution of 'sudo wget https://www.webmin.com/jcameron-key.asc', which downloads a GPG key. The output of the wget command shows the file 'jcameron-key.asc' being saved. Finally, the terminal shows the execution of 'sudo apt-key add jcameron-key.asc', which adds the downloaded key to the system's keyring.

```
administrator@plabclient01: /etc/apt$ sudo add-apt-repository "deb http://download.webmin.com/download/repository sarge contrib"
Get:1 http://security.ubuntu.com/ubuntu focal-security InRelease [114 kB]
Hit:2 http://us.archive.ubuntu.com/ubuntu focal InRelease
Ign:3 http://download.webmin.com/download/repository sarge InRelease
Get:4 http://security.ubuntu.com/ubuntu focal-security/main i386 Packages [586 kB]
Get:5 http://security.ubuntu.com/ubuntu focal-security/main amd64 Packages [2,151 kB]
Get:6 http://security.ubuntu.com/ubuntu focal-security/universe amd64 Packages [834 kB]
Get:7 http://security.ubuntu.com/ubuntu focal-security/universe i386 Packages [595 kB]
Get:8 http://download.webmin.com/download/repository sarge Release [16.9 kB]
Get:9 http://download.webmin.com/download/repository sarge Release.gpg [173 B]
Ign:9 http://download.webmin.com/download/repository sarge Release.gpg
Reading package lists... Done
W: GPG error: http://download.webmin.com/download/repository sarge Release: The following signatures couldn't be verified because the public key is not available: NO_PUBKEY D97A3AE911F63C51
E: The repository 'http://download.webmin.com/download/repository sarge Release' is not signed.
N: Updating from such a repository can't be done securely, and is therefore disabled by default.
N: See apt-get(8) manpage for repository creation and user configuration details.
administrator@plabclient01: /etc/apt$ sudo wget https://www.webmin.com/jcameron-key.asc
--2023-05-09 08:46:42-- https://www.webmin.com/jcameron-key.asc
Resolving www.webmin.com (www.webmin.com)... 216.105.38.11
Connecting to www.webmin.com (www.webmin.com)|216.105.38.11|:443... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://webmin.com/jcameron-key.asc [following]
--2023-05-09 08:46:43-- https://webmin.com/jcameron-key.asc
Resolving webmin.com (webmin.com)... 216.105.38.11
Connecting to webmin.com (webmin.com)|216.105.38.11|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1320 (1.3K) [text/plain]
Saving to: 'jcameron-key.asc'

jcameron-key.asc      100%[=====] 1.29K  --.-KB/s  in 0s

2023-05-09 08:46:43 (119 MB/s) - 'jcameron-key.asc' saved [1320/1320]

administrator@plabclient01: /etc/apt$ sudo apt-key add jcameron-key.asc
OK
administrator@plabclient01: /etc/apt$
```

Figure 2.15 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Step 4

In the **Terminal** window, type the following command and press **Enter**:

```
sudo apt update
```

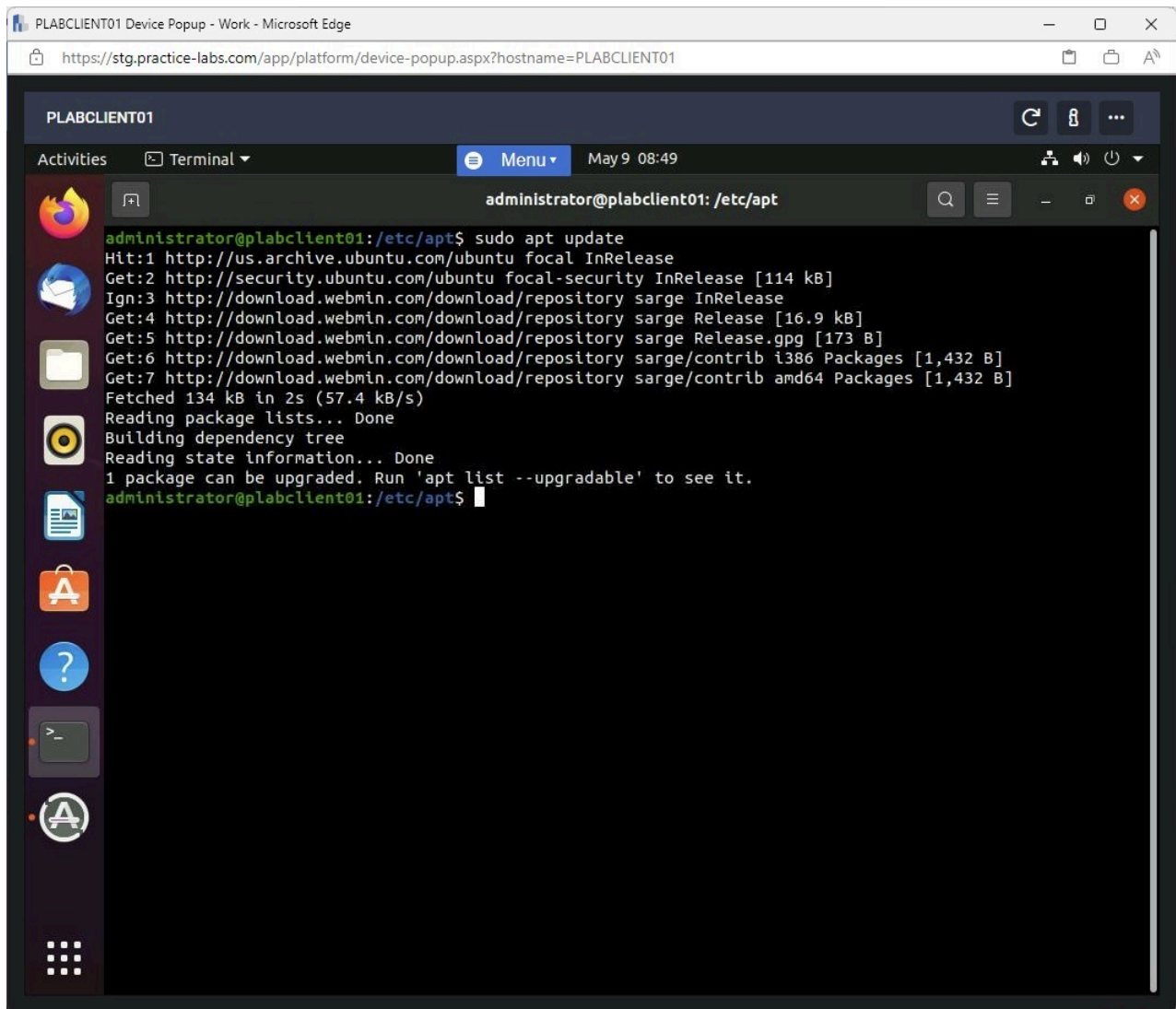



Figure 2.16 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Step 5

In the **Terminal** window, type the following command and press **Enter**:

```
sudo apt install webmin
```

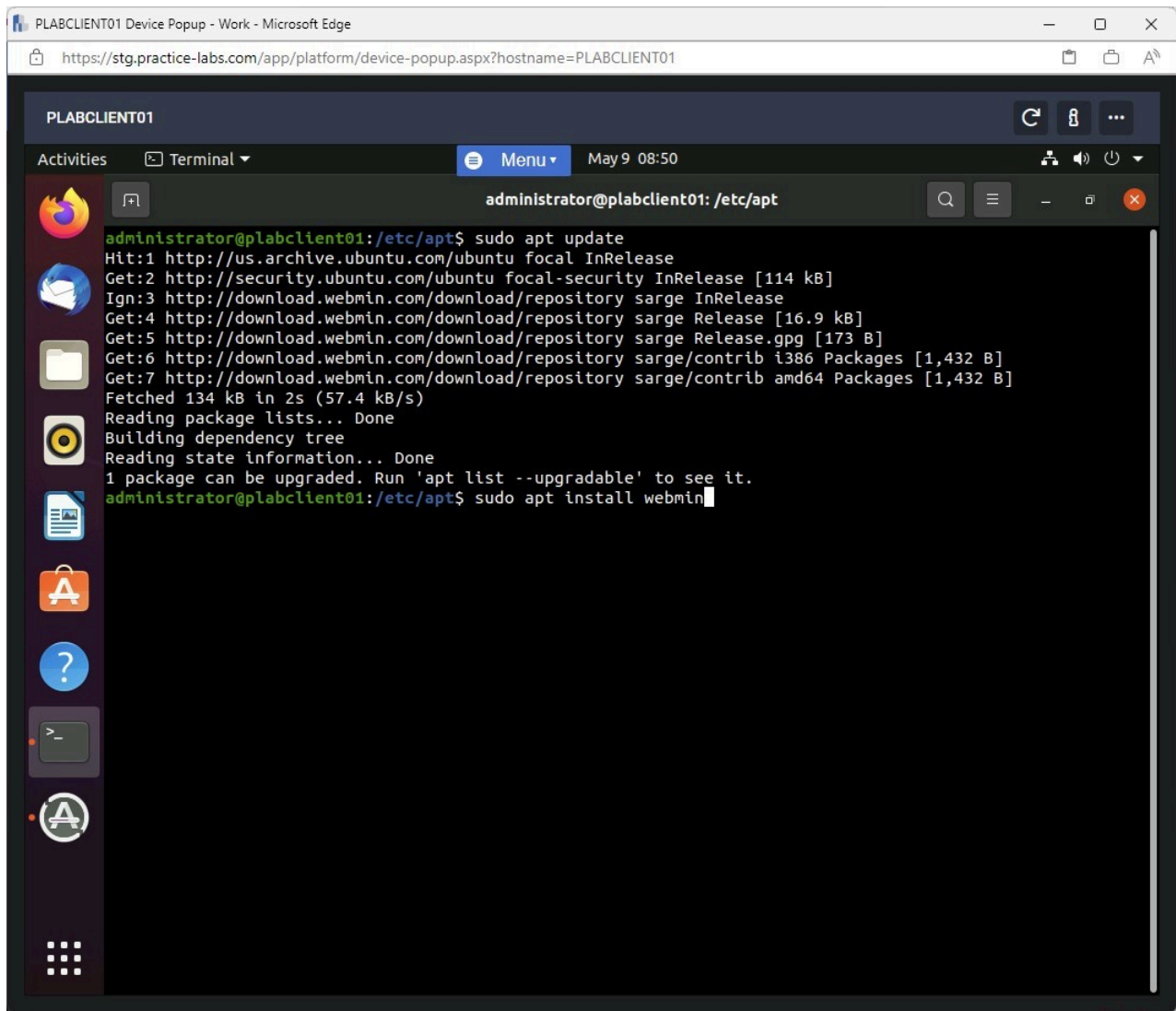


Figure 2.17 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered.

Step 6

In the **Terminal** window, type the following for the **Do you want to continue? [Y/n]** prompt:

Y

Press **Enter**.

PLABCLIENT01 Device Popup - Work - Microsoft Edge

https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABCLIENT01

PLABCLIENT01

Activities Terminal Menu May 9 08:52

administrator@plabclient01: /etc/apt

```

Reading state information... Done
The following packages were automatically installed and are no longer required:
  libassuan0:i386 libavahi-common-data:i386 libavahi-common3:i386 libffi7:i386 libkeyutils1:i386
  libnettle7:i386 libnpt0:i386 libp11-kit0:i386 libreadline8:i386 libtasn1-6:i386
  pinentry-curses:i386
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  libalgorithm-c3-perl libauthen-pam-perl libb-hooks-endofscope-perl libb-hooks-op-check-perl
  libclass-c3-perl libclass-c3-xs-perl libclass-data-inheritable-perl libclass-inspector-perl
  libclass-method-modifiers-perl libclass-singleton-perl libclass-xsaccessor-perl libdata-optlist-perl
  libdatetime-locale-perl libdatetime-perl libdatetime-timezone-perl libdevel-callchecker-perl
  libdevel-caller-perl libdevel-lexalias-perl libdevel-stacktrace-perl libdynaloader-functions-perl
  libeval-closure-perl libexception-class-perl libfile-sharedir-perl libio-pty-perl
  libmodule-implementation-perl libmodule-runtime-perl libmro-compat-perl libnamespace-autoclean-perl
  libnamespace-clean-perl libpackage-stash-perl libpackage-stash-xs-perl libpadwalker-perl
  libparams-classify-perl libparams-util-perl libparams-validationcompiler-perl libreadonly-perl
  libref-util-perl libref-util-xs-perl librole-tiny-perl libspecio-perl libsub-exporter-perl
  libsub-exporter-progressive-perl libsub-identify-perl libsub-install-perl libsub-name-perl
  libsub-quote-perl libvariable-magic-perl libxstring-perl
Suggested packages:
  libscalar-number-perl libtest-fatal-perl
The following NEW packages will be installed:
  libalgorithm-c3-perl libauthen-pam-perl libb-hooks-endofscope-perl libb-hooks-op-check-perl
  libclass-c3-perl libclass-c3-xs-perl libclass-data-inheritable-perl libclass-inspector-perl
  libclass-method-modifiers-perl libclass-singleton-perl libclass-xsaccessor-perl libdata-optlist-perl
  libdatetime-locale-perl libdatetime-perl libdatetime-timezone-perl libdevel-callchecker-perl
  libdevel-caller-perl libdevel-lexalias-perl libdevel-stacktrace-perl libdynaloader-functions-perl
  libeval-closure-perl libexception-class-perl libfile-sharedir-perl libio-pty-perl
  libmodule-implementation-perl libmodule-runtime-perl libmro-compat-perl libnamespace-autoclean-perl
  libnamespace-clean-perl libpackage-stash-perl libpackage-stash-xs-perl libpadwalker-perl
  libparams-classify-perl libparams-util-perl libparams-validationcompiler-perl libreadonly-perl
  libref-util-perl libref-util-xs-perl librole-tiny-perl libspecio-perl libsub-exporter-perl
  libsub-exporter-progressive-perl libsub-identify-perl libsub-install-perl libsub-name-perl
  libsub-quote-perl libvariable-magic-perl libxstring-perl webmin
0 upgraded, 49 newly installed, 0 to remove and 1 not upgraded.
Need to get 37.7 MB of archives.
After this operation, 221 MB of additional disk space will be used.
Do you want to continue? [Y/n] Y

```

Figure 2.18 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing Y entered.

Step 7

The webmin package is now successfully installed.

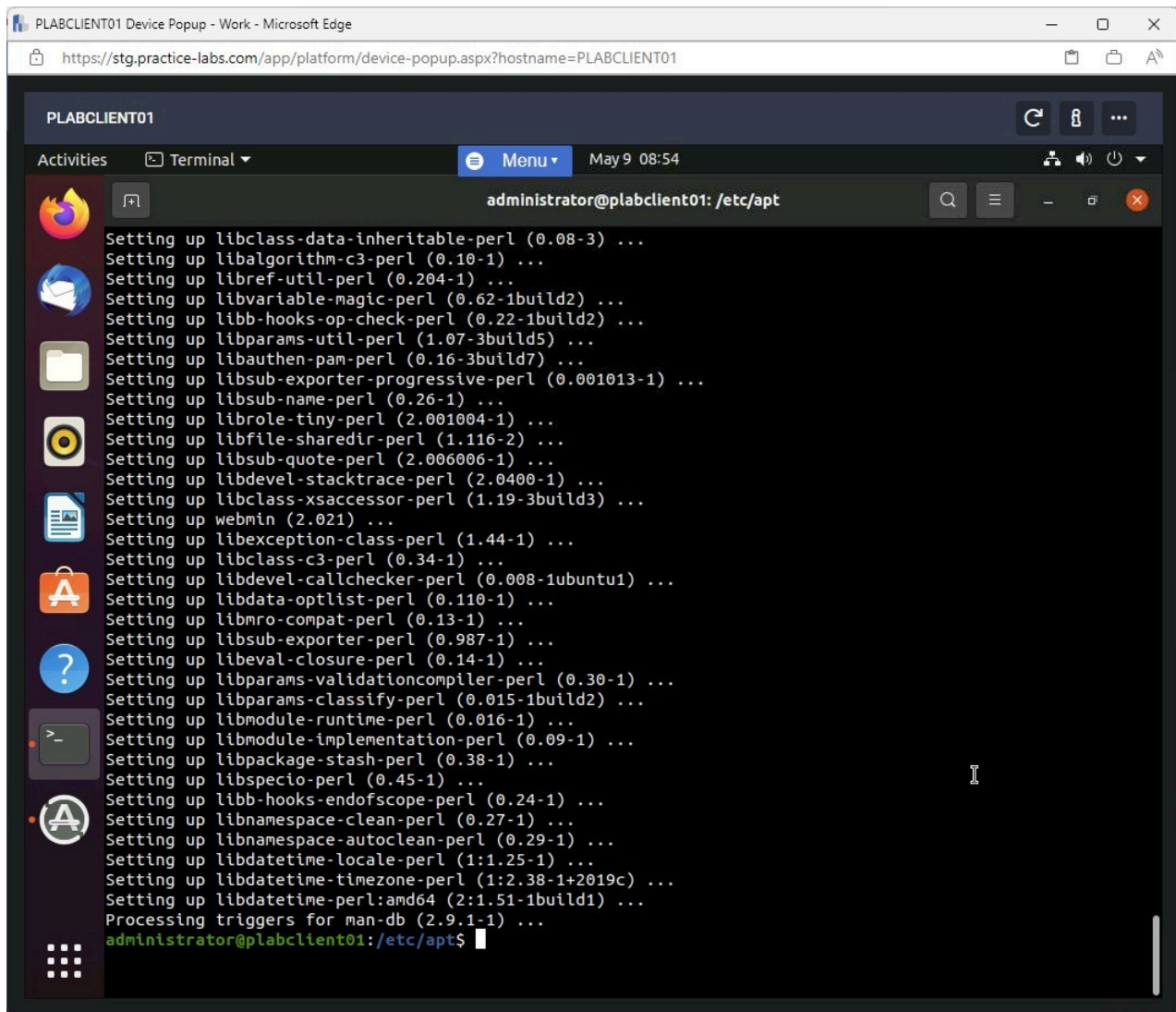


Figure 2.19 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the successful installation of the webmin package.

Keep the **Terminal** window open.

Task 3 - Upgrade and Remove Installed Software

In this task, you will use apt to upgrade and remove the installed software package and its dependencies.

Step 1

In **PLABCLIENT01**, the **Terminal** window is open.

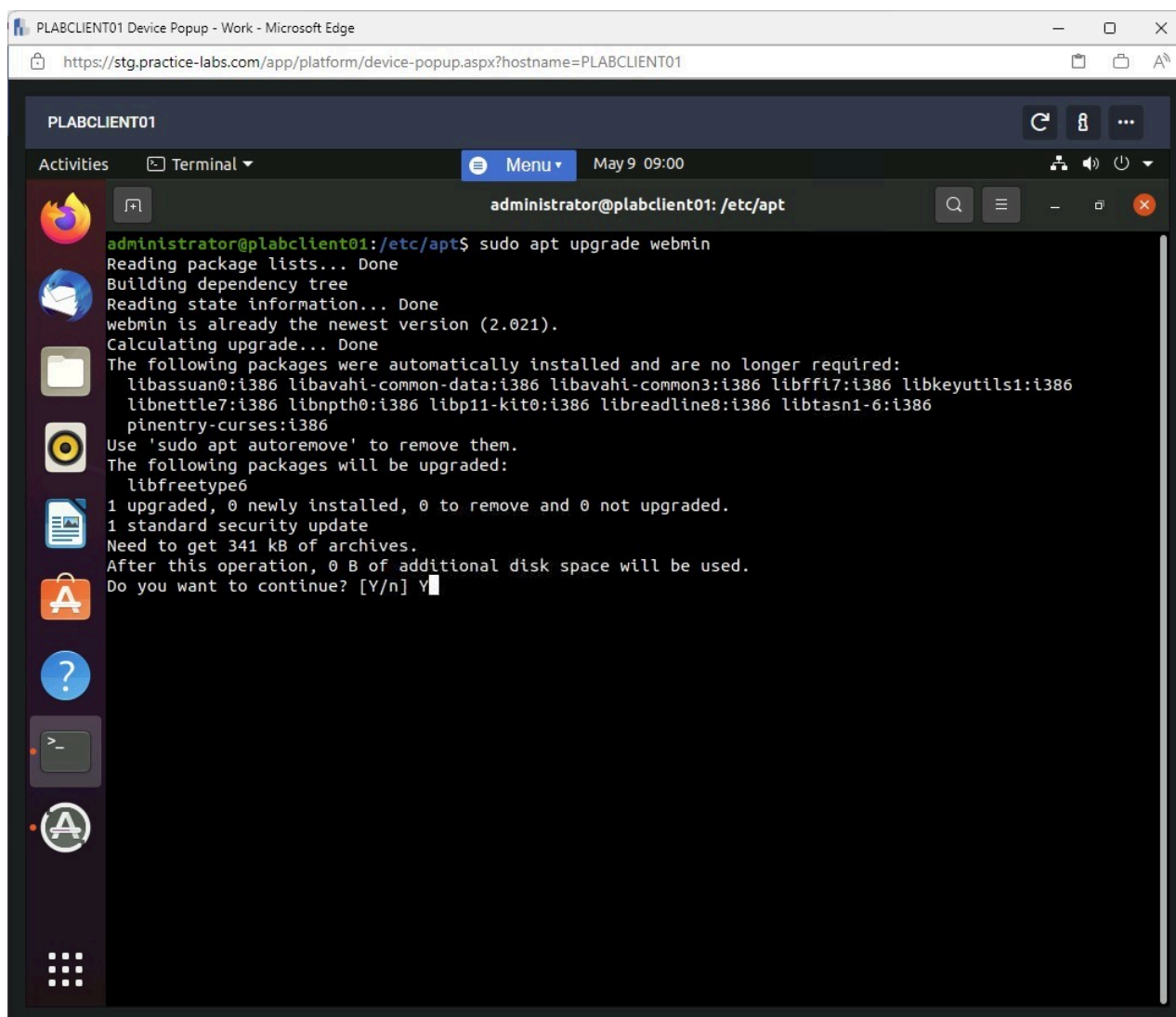
In the **Terminal** window, type the following command and press **Enter**:

```
sudo apt upgrade webmin
```


Type the following for the **Do you want to continue? [Y/n]** prompt:

Y

Press **Enter**.



```
PLABCLIENT01 Device Popup - Work - Microsoft Edge
https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABCLIENT01

PLABCLIENT01
Activities Terminal May 9 09:00
administrator@plabclient01: /etc/apt

administrator@plabclient01:/etc/apt$ sudo apt upgrade webmin
Reading package lists... Done
Building dependency tree
Reading state information... Done
webmin is already the newest version (2.021).
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
  libassuan0:i386 libavahi-common-data:i386 libavahi-common3:i386 libffi7:i386 libkeyutils1:i386
  libnettle7:i386 libnpt0:i386 libp11-kit0:i386 libreadline8:i386 libtasn1-6:i386
  pinentry-curses:i386
Use 'sudo apt autoremove' to remove them.
The following packages will be upgraded:
  libfreetype6
1 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
1 standard security update
Need to get 341 kB of archives.
After this operation, 0 B of additional disk space will be used.
Do you want to continue? [Y/n] Y
```

Figure 2.20 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered.

Step 2

The **webmin** package is now successfully upgraded.

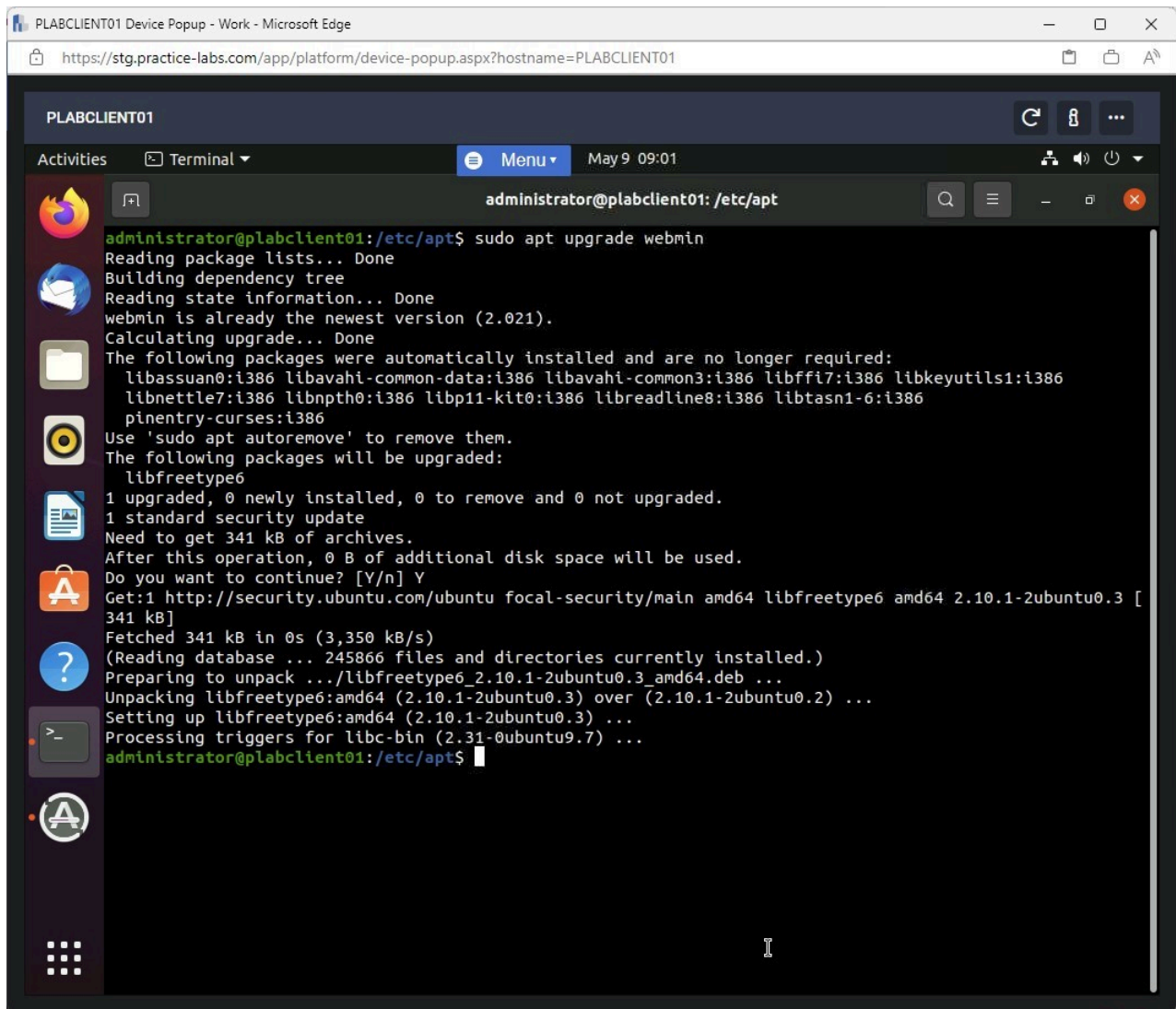


Figure 2.21 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the successful upgrade of the webmin package.

Step 3

In the **Terminal** window, type the following command and press **Enter**:

```
sudo apt remove webmin
```

Type the following for the **Do you want to continue? [Y/n]** prompt:

```
Y
```

Press **Enter**.

The screenshot shows a terminal window titled 'PLABCLIENT01 Device Popup - Work - Microsoft Edge'. The address bar shows the URL 'https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABCLIENT01'. The terminal window has a top bar with 'PLABCLIENT01', a 'Menu' button, and the date 'May 9 09:04'. The terminal prompt is 'administrator@plabclient01: /etc/apt'. The output shows the following commands and their results:

```

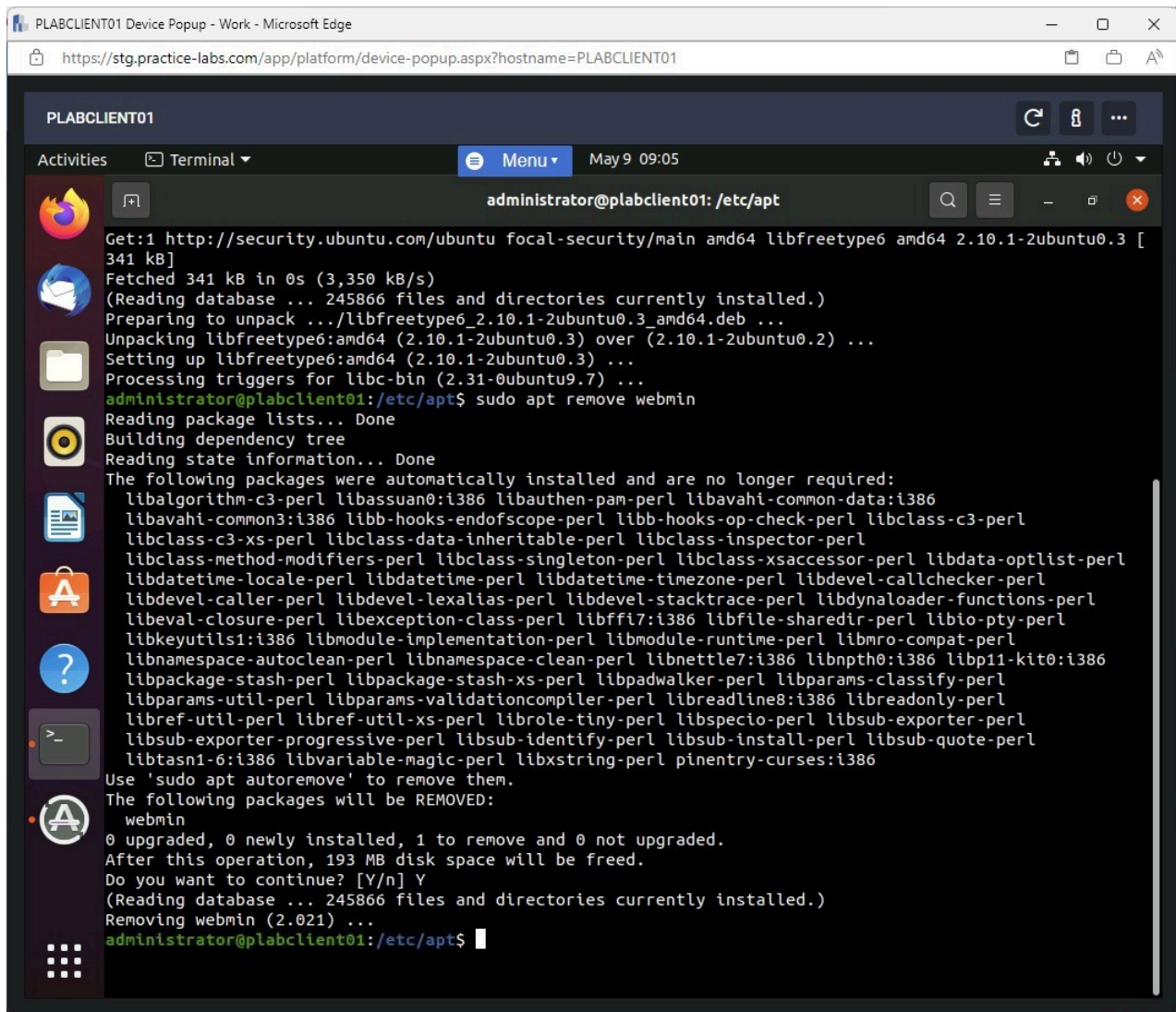
1 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
1 standard security update
Need to get 341 kB of archives.
After this operation, 0 B of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 http://security.ubuntu.com/ubuntu focal-security/main amd64 libfreetype6 amd64 2.10.1-2ubuntu0.3 [
341 kB]
Fetched 341 kB in 0s (3,350 kB/s)
(Reading database ... 245866 files and directories currently installed.)
Preparing to unpack .../libfreetype6_2.10.1-2ubuntu0.3_amd64.deb ...
Unpacking libfreetype6:amd64 (2.10.1-2ubuntu0.3) over (2.10.1-2ubuntu0.2) ...
Setting up libfreetype6:amd64 (2.10.1-2ubuntu0.3) ...
Processing triggers for libc-bin (2.31-0ubuntu9.7) ...
administrator@plabclient01:/etc/apt$ sudo apt remove webmin
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following packages were automatically installed and are no longer required:
libalgorithm-c3-perl libassuan0:i386 libauthen-pam-perl libavahi-common-data:i386
libavahi-common3:i386 libb-hooks-endofscope-perl libb-hooks-op-check-perl libclass-c3-perl
libclass-c3-xs-perl libclass-data-inheritable-perl libclass-inspector-perl
libclass-method-modifiers-perl libclass-singleton-perl libclass-xsaccessor-perl libdata-optlist-perl
libdatetime-locale-perl libdatetime-perl libdatetime-timezone-perl libdevel-callchecker-perl
libdevel-caller-perl libdevel-lexalias-perl libdevel-stacktrace-perl libdynaloader-functions-perl
libdevel-closure-perl libexception-class-perl libffi7:i386 libfile-sharedir-perl libio-pty-perl
libkeyutils1:i386 libmodule-implementation-perl libmodule-runtime-perl libmro-compat-perl
libnamespace-autoclean-perl libnamespace-clean-perl libnettle7:i386 libnpt0:i386 libp11-kit0:i386
libpackage-stash-perl libpackage-stash-xs-perl libpadwalker-perl libparams-classify-perl
libparams-util-perl libparams-validationcompiler-perl libreadline8:i386 libreadonly-perl
libref-util-perl libref-util-xs-perl librole-tiny-perl libspecio-perl libsub-exporter-perl
libsub-exporter-progressive-perl libsub-identify-perl libsub-install-perl libsub-quote-perl
libtasn1-6:i386 libvariable-magic-perl libxstring-perl pinentry-curses:i386
Use 'sudo apt autoremove' to remove them.
The following packages will be REMOVED:
webmin
0 upgraded, 0 newly installed, 1 to remove and 0 not upgraded.
After this operation, 193 MB disk space will be freed.
Do you want to continue? [Y/n] Y

```

Figure 2.22 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered.

Step 4

The **webmin** software package is now removed.



The screenshot shows a terminal window titled 'PLABCLIENT01 Device Popup - Work - Microsoft Edge'. The address bar shows the URL 'https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABCLIENT01'. The terminal window has a title bar 'PLABCLIENT01' and a menu bar with 'Activities', 'Terminal', and 'Menu'. The terminal prompt is 'administrator@plabclient01: /etc/apt'. The output shows the installation of libfreetype6 and the removal of webmin. The terminal text is as follows:

```
Get:1 http://security.ubuntu.com/ubuntu focal-security/main amd64 libfreetype6 amd64 2.10.1-2ubuntu0.3 [341 kB]
Fetched 341 kB in 0s (3,350 kB/s)
(Reading database ... 245866 files and directories currently installed.)
Preparing to unpack .../libfreetype6_2.10.1-2ubuntu0.3_amd64.deb ...
Unpacking libfreetype6:amd64 (2.10.1-2ubuntu0.3) over (2.10.1-2ubuntu0.2) ...
Setting up libfreetype6:amd64 (2.10.1-2ubuntu0.3) ...
Processing triggers for libc-bin (2.31-0ubuntu9.7) ...
administrator@plabclient01:/etc/apt$ sudo apt remove webmin
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following packages were automatically installed and are no longer required:
 libalgorithm-c3-perl libassuan0:i386 libauthen-pam-perl libavahi-common-data:i386
 libavahi-common3:i386 libb-hooks-endofscope-perl libb-hooks-op-check-perl libclass-c3-perl
 libclass-c3-xs-perl libclass-data-inheritable-perl libclass-inspector-perl
 libclass-method-modifiers-perl libclass-singleton-perl libclass-xsaccessor-perl libdata-optlist-perl
 libdatetime-locale-perl libdatetime-perl libdatetime-timezone-perl libdevel-callchecker-perl
 libdevel-caller-perl libdevel-lexalias-perl libdevel-stacktrace-perl libdynaloader-functions-perl
 libeval-closure-perl libexception-class-perl libffi7:i386 libfile-sharedir-perl libio-pty-perl
 libkeyutils1:i386 libmodule-implementation-perl libmodule-runtime-perl libmro-compat-perl
 libnamespace-autoclean-perl libnamespace-clean-perl libnettle7:i386 libnpt0:i386 libp11-kit0:i386
 libpackage-stash-perl libpackage-stash-xs-perl libpadwalker-perl libparams-classify-perl
 libparams-util-perl libparams-validationcompiler-perl libreadline8:i386 libreadonly-perl
 libref-util-perl libref-util-xs-perl librole-tiny-perl libspecio-perl libsub-exporter-perl
 libsub-exporter-progressive-perl libsub-identify-perl libsub-install-perl libsub-quote-perl
 libtasn1-6:i386 libvariable-magic-perl libxstring-perl pinentry-curses:i386
Use 'sudo apt autoremove' to remove them.
The following packages will be REMOVED:
 webmin
0 upgraded, 0 newly installed, 1 to remove and 0 not upgraded.
After this operation, 193 MB disk space will be freed.
Do you want to continue? [Y/n] Y
(Reading database ... 245866 files and directories currently installed.)
Removing webmin (2.021) ...
administrator@plabclient01:/etc/apt$
```

Figure 2.23 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the successful removal of the webmin package.

Step 5

In the **Terminal** window, type the following command and press **Enter**:

```
sudo apt autoremove
```

Type the following for the **Do you want to continue? [Y/n]** prompt:

```
Y
```

Press **Enter**.

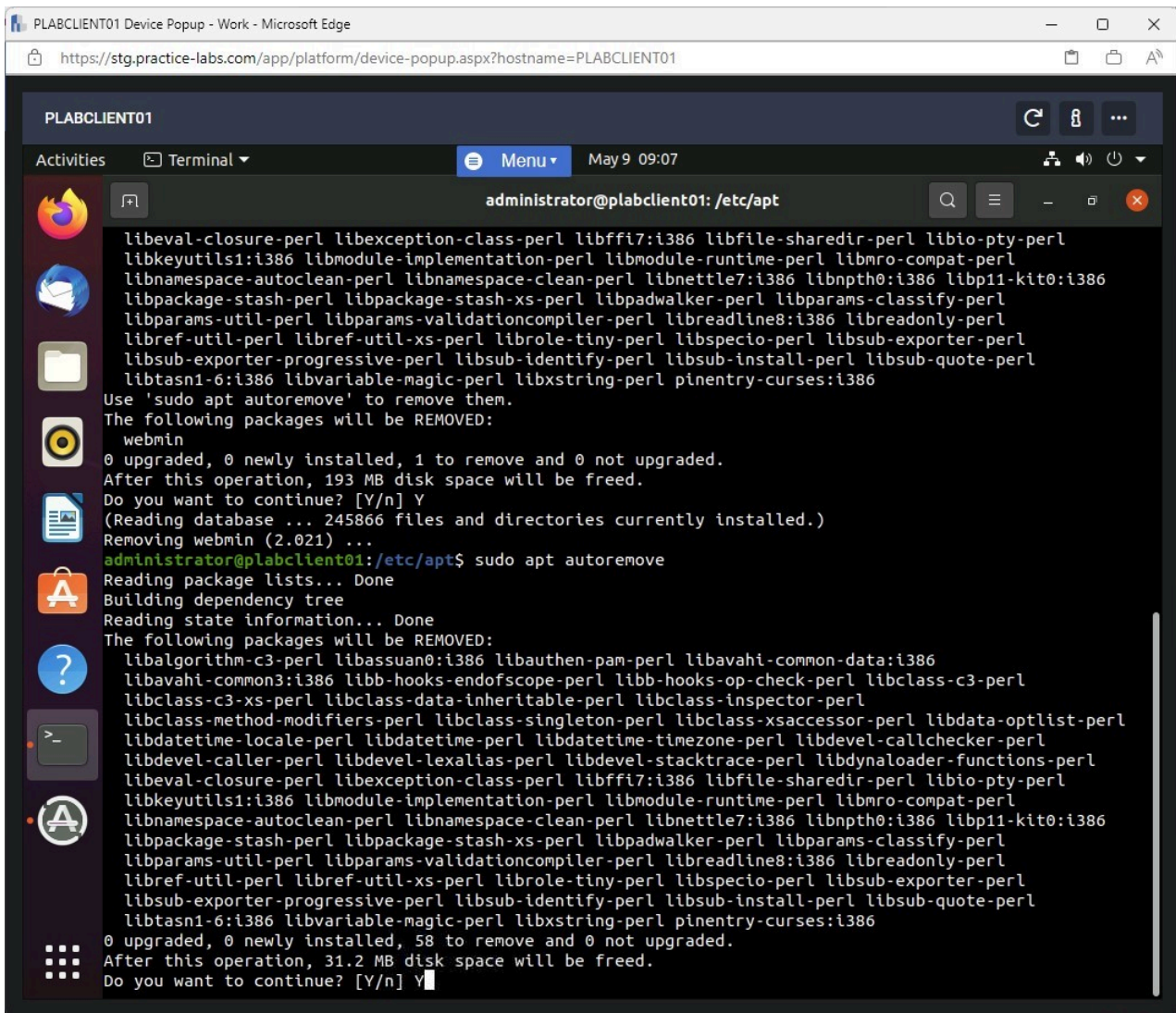
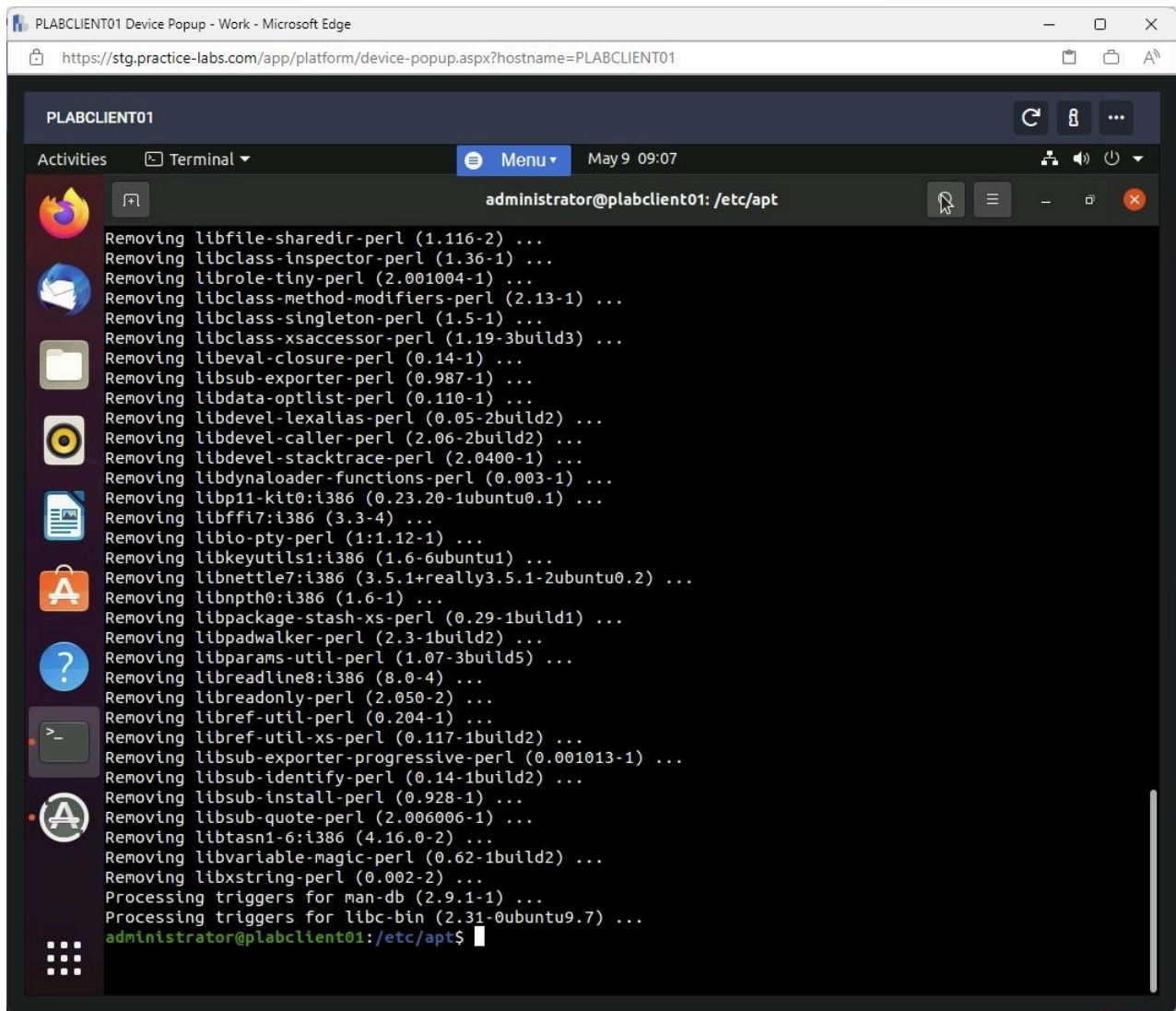


Figure 2.24 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered.

Step 6

The **autoremove** option will remove the orphaned dependencies.



The screenshot shows a terminal window titled 'PLABCLIENT01' with the URL 'https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABCLIENT01'. The terminal displays the command 'apt autoremove' and its output, which lists various Perl modules and other packages being removed. The output includes:

```
Removing libfile-sharedir-perl (1.116-2) ...
Removing libclass-inspector-perl (1.36-1) ...
Removing librole-tiny-perl (2.001004-1) ...
Removing libclass-method-modifiers-perl (2.13-1) ...
Removing libclass-singleton-perl (1.5-1) ...
Removing libclass-xsaccessor-perl (1.19-3build3) ...
Removing libeval-closure-perl (0.14-1) ...
Removing libsub-exporter-perl (0.987-1) ...
Removing libdata-optlist-perl (0.110-1) ...
Removing libdevel-lexalias-perl (0.05-2build2) ...
Removing libdevel-caller-perl (2.06-2build2) ...
Removing libdevel-stacktrace-perl (2.0400-1) ...
Removing libdynaloader-functions-perl (0.003-1) ...
Removing libp11-kit0:i386 (0.23.20-1ubuntu0.1) ...
Removing libffi7:i386 (3.3-4) ...
Removing libio-pty-perl (1:1.12-1) ...
Removing libkeyutils1:i386 (1.6-6ubuntu1) ...
Removing libnettle7:i386 (3.5.1+really3.5.1-2ubuntu0.2) ...
Removing libnpt0:i386 (1.6-1) ...
Removing libpackage-stash-xs-perl (0.29-1build1) ...
Removing libpadwalker-perl (2.3-1build2) ...
Removing libparams-util-perl (1.07-3build5) ...
Removing libreadline8:i386 (8.0-4) ...
Removing libreadonly-perl (2.050-2) ...
Removing libref-util-perl (0.204-1) ...
Removing libref-util-xs-perl (0.117-1build2) ...
Removing libsub-exporter-progressive-perl (0.001013-1) ...
Removing libsub-identify-perl (0.14-1build2) ...
Removing libsub-install-perl (0.928-1) ...
Removing libsub-quote-perl (2.006006-1) ...
Removing libtasn1-6:i386 (4.16.0-2) ...
Removing libvariable-magic-perl (0.62-1build2) ...
Removing libxstring-perl (0.002-2) ...
Processing triggers for man-db (2.9.1-1) ...
Processing triggers for libc-bin (2.31-0ubuntu9.7) ...
administrator@plabclient01:/etc/apt$
```

Figure 2.25 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the successful removal of the orphaned dependencies.

Close the **Terminal** window.

Keep all devices that you have powered on in their current state and proceed to the review section.

Review

Well done, you have completed the **Package Management and Updating Linux Devices** Practice Lab.

Summary

You completed the following exercises:

- Exercise 1 - Install and Update Software using Dnf
- Exercise 2 - Install and Update Software using Apt

You should now be able to:

- Use Dnf to Explore and Install a Current Repository
- Install a Custom Repository
- Remove an Installed Software Package
- Use Apt to Explore and Install a Current Repository
- Install a Third-party Repository
- Upgrade and Remove Installed Software

Feedback

Shutdown all virtual machines used in this lab. Alternatively, you can log out of the lab platform.